

An expanded palette of bright and photostable organellar Ca^{2+} sensors

Reviewed Preprint

v1 • August 26, 2025

Not revised

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eLife Assessment

This **important** study introduces a new class of spectrally tunable, dye-based calcium sensors optimized for imaging in organelles with high calcium concentrations, such as the endoplasmic reticulum and mitochondria. The experimental evidence supporting the applicability of these sensors is **convincing**, with thorough validation in cultured cells and neurons. The work will be of high interest to researchers studying calcium signaling dynamics in subcellular compartments.

<https://doi.org/10.7554/eLife.107845.1.sa2>

Abstract

The use of fluorescent sensors for functional imaging has revolutionized the study of organellar Ca^{2+} signaling. However, understanding the dynamic interplay between intracellular Ca^{2+} sinks and sources has been hindered by the lack of bright, photostable, and multiplexed measurements in different organelles, limiting our ability to define how Ca^{2+} shapes cell physiology across fields of biology. Here we introduce a new toolkit of chemigenetic organellar Ca^{2+} indicators whose color is tunable by reconstituting their fluorescence with different exogenous rhodamine dye-ligands, which significantly expand the capacity for multiplexing organellar Ca^{2+} measurements. These sensors, which we named ER-HaloCaMP and Mito-HaloCaMP, are optimized to report Ca^{2+} dynamics in the endoplasmic reticulum (ER) and mitochondria of mammalian cells and neurons, and show significantly improved brightness, photostability and responsiveness when compared to current best-in-class alternatives. Using either red or far-red dye-ligands, both ER-HaloCaMP and Mito-HaloCaMP enable visualizing ER and mitochondrial Ca^{2+} dynamics in neuronal axons, a subcellular location that only contains a few ER tubules and small mitochondria, structural limitations that have impaired measurements with previous red sensors. To show the expanded multiplexing capacities of our toolkit, we measured interorganellar Ca^{2+} fluxes simultaneously in three different subcellular compartments in live cells, revealing that the amplitude of ER Ca^{2+} release controls the efficacy of ER-mitochondria Ca^{2+} coupling in a cooperative manner. Organellar HaloCaMPs enable also measuring Ca^{2+} dynamics in intact brain tissue from flies and rodents, demonstrating their versatility across biological models.

Our new toolkit provides an expanded palette of bright, photostable and responsive organellar Ca^{2+} sensors, which will facilitate future studies of intracellular Ca^{2+} signaling across fields of biology in health and disease.

Introduction

The ability of cells to generate transient Ca^{2+} fluctuations in subcellular locales is fundamental for controlling essential cellular processes such as gene expression, cell metabolism or neuronal communication^{1,2}. Organelles such as mitochondria or the ER can function within cells as sinks or sources of Ca^{2+} . They modulate Ca^{2+} dynamics across cellular compartments with temporal and spatial precision, shaping cell physiology^{3,4}. For ample, Ca^{2+} transfer between the ER and mitochondria accelerates mitochondrial metabolism to fuel energy-intensive tasks⁵, modulates mitochondrial fission-fusion dynamics^{6,7} or influences central cellular processes such as autophagy and stress responses^{8,9}. On the other hand, mitochondria and ER uptake Ca^{2+} in neurons during neurotransmission, shaping the capacity of neuronal communication¹⁰⁻¹⁴. Not surprisingly, owing to the critical role of ER and mitochondrial Ca^{2+} signaling in both physiological and pathological processes^{3,15-17}, their investigation has garnered significant attention over the past several decades^{2,18}.

For many years, insights into organellar Ca^{2+} functions were primarily obtained using indirect cytosolic Ca^{2+} measurements coupled with organelle manipulations'. However, recent advancements now enable measuring Ca^{2+} dynamics directly within the ER or mitochondria using adapted genetically encoded sensors (GECs)^{11,12,19-22}. Among previous mitochondria and ER Ca^{2+} sensors, GCaMP variants have stood out due to their high dynamic range and sensitivity^{11-14,19-23,24}. However, for intraorganellar use they have inherent limitations. First, their excitation spectrum overlaps with other commonly used fluorescent proteins, sensors and optogenetic tools²⁵⁻²¹, complicating multicolor imaging needed for dissecting complex cellular processes; second, blue light excites autofluorescence in biological tissues²⁸, making it difficult to detect calcium dynamics in small organellar structures with green fluorescent indicators^{12,22}; third, imaging with blue light is associated with higher phototoxicity^{29,30}, which may complicate measurements over time. These limitations suggest that red-shifted GECs could provide a reliable imaging alternative for imaging Ca^{2+} dynamics in organelles. However, despite recent advances, available red-shifted GECs for ER and mitochondrial Ca^{2+} imaging are not ideal due to either an inappropriately high Ca^{2+} affinity^{2,19} or a limited dynamic range^{19,20}. Additionally, the lack of far-red Ca^{2+} sensors for these organelles has further limited simultaneous multi-wavelength imaging of organellar and cytosolic Ca^{2+} dynamics, along with other key physiological markers, limiting multi-compartment dissections of Ca^{2+} fluxes and their relevance for cellular physiology in health and disease.

Leveraging HaloTag³¹ as a scaffold for generating novel sensors has provided several novel indicators that enable measuring diverse pieces of biology in different colors, as the color of emission is provided by different membrane-permeable rhodamine dye-ligands that irreversibly bind to the HaloTag. These include sensors for voltage^{32,33}, ATP³⁴, NAD³⁴, dopamine³⁵, pH³⁶ and protein aggregation³⁷, among others^{38,39}. However, while several HaloTag-based strategies have also provided robust multicolor cytosolic Ca^{2+} sensors⁴⁰⁻⁴³, suitable versions to expand the palette of organellar Ca^{2+} sensors do not exist. In this work, we present a new suite of chemogenetic indicators derived from HaloCaMP, a genetically encoded Ca^{2+} sensor whose fluorescence is provided by membrane-permeable Janelia Fluor rhodamine dye-ligands with a choice of colors appended to the HaloTag Ligand (JF-HTL)⁴⁰. We generated novel organellar HaloCaMP sensors, tailored specifically for Ca^{2+} measurements in ER and mitochondria: ER-HaloCaMP and Mito-HaloCaMP. These sensors combine robust responsiveness with the enhanced brightness and photostability provided by JF-HTLs. Along with these advantages, the indicators feature tunable red and far-red emission, enabling multicolor imaging

without the spectral overlap or interference issues commonly associated with green GECIs. We compared ER-HaloCaMP and Mito-HaloCaMP to existing red-emitting organellar sensors, and found they responded significantly better (1.6-to 2.3-fold) and were significantly brighter (10-to 50-fold). Physiological responses of ER-HaloCaMP reconstituted with JF₅₈₅-HTL (ER-HaloCaMP₅₈₅) in cells and neurons were indistinguishable from the those of the gold-standard green ER Ca²⁺ sensor ER-GCaMP6-210¹², but the photostability and brightness-over-background of ER-HaloCaMP₅₈₅ were significantly improved, making it the ideal sensor for ER Ca²⁺ measurements. This work provides a new technological framework to enable novel investigations on cellular Ca²⁺ fluxes, offering new insights into the fundamental role of Ca²⁺ signaling in health and disease.

Results

Developing a bright and highly responsive

red ER Ca²⁺ sensor based on HaloCaMP

The ER contains ~2000 times more Ca²⁺ than the cytosol. This surplus of Ca²⁺ inevitably saturates all existing red cytosolic Ca²⁺ GECIs^{44,49}, making them unresponsive to changes in ER Ca²⁺. To overcome this issue and image ER Ca²⁺ in red, previous efforts have screened for mutations that lower the affinity of red GECI to match resting ER Ca²⁺ levels. These approaches generated LAR-GECOs, which preserved a good dynamic range yet their affinity for Ca²⁺ was not lowered enough to match ER Ca²⁺ levels, limiting their usability²¹. Alternative efforts generated R-CEPIAer, which presented low affinity but its responsiveness remained ten times lower compared to that of green ER Ca²⁺ GECIs¹⁹.

To overcome the limitations of existing red ER Ca²⁺ sensors and expand their spectral range beyond red, we first generated a series of HaloCaMP1a variants harboring mutations in key Ca²⁺-coordinating sites to lower their affinity (Fig. 1A; Supp. Fig. S1A; Table 1). We then performed Ca²⁺ titrations and identified a variant that presented both high dynamic range (~15 ΔF/F₀) and low affinity (EC₅₀ = 86 μM), which we named Low Affinity (LA)-HaloCaMP₅₈₅ (Fig. 1B). The general properties of this low affinity variant did not significantly change when Ca²⁺ titrations were performed at 37 °C, supporting its usability for live cell experiments (Supp. Fig. S1C; EC₅₀ = 89 μM, ~17 ΔF/F₀). As a control, we confirmed that micromolar/millimolar increases in Ca²⁺ did not cause any increase in fluorescence of HaloTag reconstituted by JF₅₈₅-HTL (HaloTag₈₅; Fig. 1B). Using purified LA-HaloCaMP₅₈₅, we confirmed that its excitation and emission spectra yielded a peak excitation at 593 nm and emission at 607 nm, confirming its usability as a red sensor (Supp. Fig. S1D).

We next targeted LA-HaloCaMP to the lumen of the ER by flanking it with a modified calreticulin signal peptide and a KDEL retention motif, as successfully used for ER-GCaMP6¹². This construct, which also was fused to mTagBFP2, was efficiently expressed in the ER of HeLa cells. We confirmed that JF₅₈₅-HTL successfully traversed both the plasma and ER membranes, demonstrating that conjugation between sensor and dye-ligand in the ER is possible (Fig. 1C). We performed *in-cell* Ca²⁺ titrations in permeabilized HEK cells and found that ER-HaloCaMP presents an affinity of 115 μM in cells (Supp. Fig. S1D), slightly lower than purified protein. Such a shift between purified and *in-cell* sensor properties is expected and has been observed for ER-GCaMP6 variants⁵⁰. Given the higher dynamic range of ER-HaloCaMP and its better suited affinity for ER Ca²⁺ than LAR-GECO and R-CEPIAer in purified protein (Fig. 1D), we next sought to compare its responsiveness against these sensors under common ER Ca²⁺ perturbations, such as thapsigargin-induced ER Ca²⁺ depletion⁵¹. Quantifying responses in both HeLa cells and somata of primary cortical neurons, we observed that although LAR-GECO and R-CEPIAer responded to thapsigargin as previously reported^{19,21,52}, ER-HaloCaMP₅₈₅ exhibited a significantly larger response in both cell types (Fig. 1E).

Table 1.

Properties of low affinity variants measured in purified protein at room temperature

Construct	EC50 (μM) <i>JF₅₈₅</i> -HTL	Dynamic range <i>JF₅₈₅</i> -HTL	Hill coefficient <i>JF₅₈₅</i> -HTL	EC50 (μM) <i>JF₆₃₅</i> -HTL	Dynamic range <i>JF₆₃₅</i> -HTL	Hill coefficient <i>JF₆₃₅</i> -HTL	Mutations
LA-HaloCaMP	86	15	1.9	34	7.2	1.4	A354E, D361G, D397G, N399D, D417Y, T418R, S420T, R429G
LA-HaloCaMP1.1	133	18	2.1	57	8.2	1.6	D361G, D397G
LA-HaloCaMP1.2	160	11	2.1	76	6.3	1.6	D361G, D397G, D434G
LA-HaloCaMP1.3	336	6	2.3	415	7.7	1.3	A354E, D361G, D397G
LA-HaloCaMP1.4	97	6	2.0	34	3.8	1.7	A354E, D361G, D397G, N399D, D417Y, T418R, S420T, R429G, D434G
LA-HaloCaMP1.5	294	3	3.0	55	2.8	1.7	A354E, D361G, D397G, N399D, D417Y, T418R, S420T, R429G, D434G, D472G
Construct	EC50 (μM)		Dynamic range		Hill coefficient		Mutations
GCaMP6-150	108		30		1.9		Previously described ¹²

Table 2.

Biophysical properties of purified LA-HaloCaMP in comparison to previous sensors.

LA-HaloCaMP	Abs _{apo} (nm)	Abs _{sat} (nm)	Ex/Em _{apo} (nm)	Ex/Em _{sat} (nm)	ϵ_{apo} (x1000) ($\text{M}^{-1} \text{cm}^{-1}$)	ϵ_{sat} (x1000) ($\text{M}^{-1} \text{cm}^{-1}$)	Φ_{apo}	Φ_{sat}	Brightness Apo ($\text{mM}^{-1} \text{cm}^{-1}$)	Brightness Sat ($\text{mM}^{-1} \text{cm}^{-1}$)
<i>JF₅₈₅</i> -HTL	591	592	593/609	593/607	8	129	N.D.	0.70	N.D.	90
<i>JF₆₃₅</i> -HTL	642	641	643/655	641/654	15	74	N.D.	0.65	N.D.	48
GECIs	Abs _{apo} (nm)	Abs _{sat} (nm)	Ex/Em _{apo} (nm)	Ex/Em _{sat} (nm)	ϵ_{apo} (x1000) ($\text{M}^{-1} \text{cm}^{-1}$)	ϵ_{sat} (x1000) ($\text{M}^{-1} \text{cm}^{-1}$)	Φ_{apo}	Φ_{sat}	Brightness Apo ($\text{mM}^{-1} \text{cm}^{-1}$)	Brightness Sat ($\text{mM}^{-1} \text{cm}^{-1}$)
LAR-GECO1	574	561	598	589	5	36	0.13	0.2	0.7	7.2
R-CEPIA1 _{er}	576	562	593	584	5	35	0.08	0.18	0.5	6.2

N.D. = not determined. The photon output of the sensor was too low to accurately determine quantum yield in the HTL = HaloTag ligand. apo = in presence of NTA. sat = in presence of saturating Ca^{2+} . Ex = excitation maximum. Em = emission maximum. ϵ = molar extinction coefficient. Φ = quantum yield. Brightness ($\text{mM}^{-1} \text{cm}^{-1}$) = the product of molar extinction coefficient $\times$ quantum yield divided by 1000.

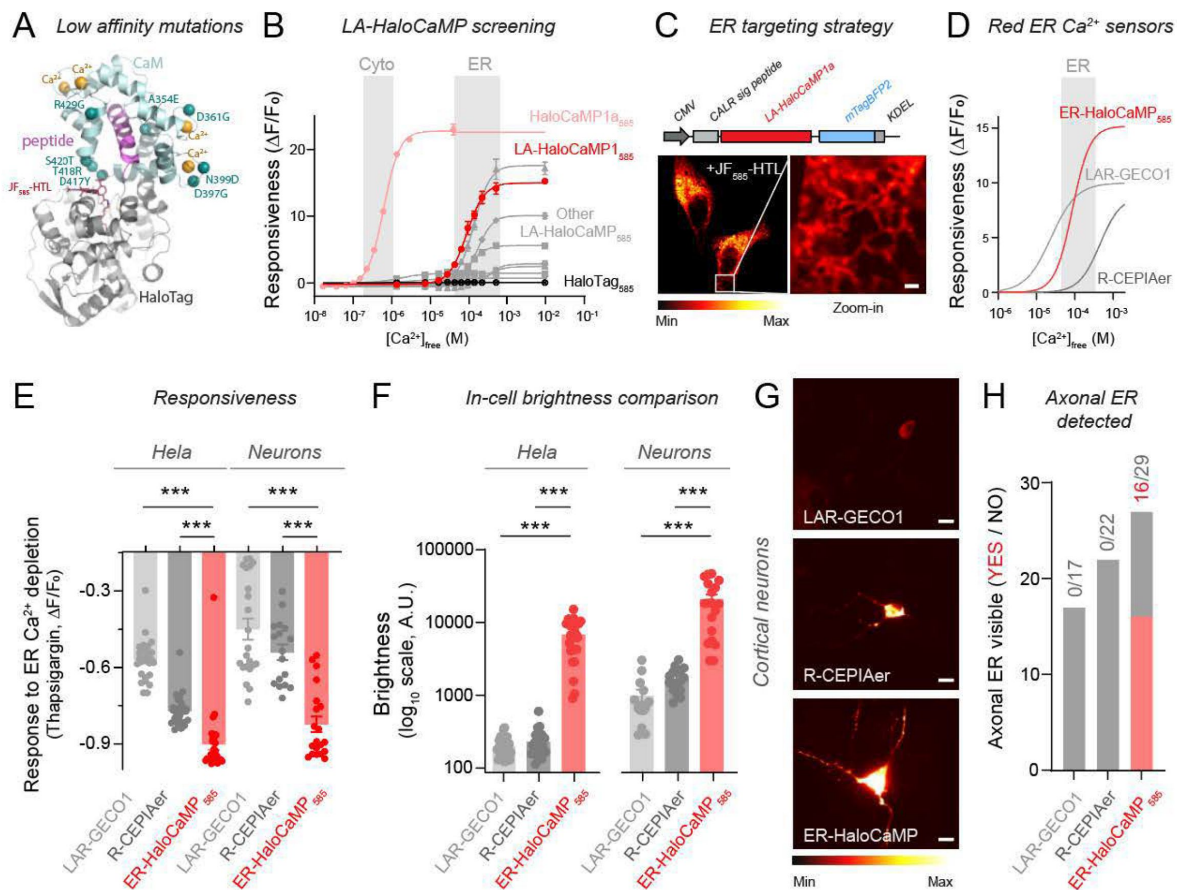


Figure 1.

ER-HaloCaMP₅₈₅ is a bright and highly responsive red ER Ca²⁺ sensor.

(A) Chai Discovery-predicted structure of low affinity HaloCaMP showing the mutations in the CaM domain in turquoise, Ca²⁺ binding sites in yellow and JF₅₈₅-HTL in red. The HaloTag is shown in grey. (B) Screening for low affinity HaloCaMP variants: *in vitro* Ca²⁺ titrations of different purified HaloCaMP variants conjugated with JF₅₈₅-HTL. (C) Top: targeting scheme for expression in the ER by adding the N-terminal signal peptide of calreticulin (CALR sig peptide) and the C-terminal KDEL retention motif. Bottom: high resolution image of a HeLa cell expressing ER-localized LA-HaloCaMP reconstituted with JF₅₈₅-HTL shows the ER structure. Pseudocolor scale shows low to high fluorescence intensity. Scale bar, 1 μ m. (D) Comparative responsiveness of ER-HaloCaMP₅₈₅, LAR-GECO1 and R-CEPIAer. (E) Quantification of thapsigargin response (1 μ M) in both HeLa cells and cortical neuron somas. (F-H) Comparison of relative brightness of LAR-GECO1, R-CEPIAer and ER-HaloCaMP₅₈₅ using identical illumination and detection conditions in either HeLa cells or cortical neurons. (F) Quantification of sensor brightness in intact HeLa cells or cortical neuron somata. (G) Representative images of brightness level in neuronal somata. Scale bar, 10 μ m. (H) Quantification of the ability to detect ER tubules in axons using red ER calcium sensors. Data are represented as mean $\pm$ SEM. See Supplementary Table ST1 for details on statistical tests and sample sizes.

We next compared the brightness of ER-HaloCaMP₅₈₅ against previous red ER Ca²⁺ sensors. While the ER is abundant and easily detectable when measured in bulk within a cell, the small ~60 nm diameter of individual ER tubules can hinder ER Ca²⁺ detection in subcellular locales. This problem becomes apparent when studying neuronal projections, as only a single narrow ER tubule may be present in axons^{53,54} or dendritic spines^{54,55}. We thus reasoned that sensors with increased brightness should facilitate studying ER Ca²⁺ in subcellular compartments in neurons and other cells. We first compared the molecular brightness in the Ca²⁺-saturated state of ER-HaloCaMP₅₈₅ against previous red ER Ca²⁺ sensors. Purified ER-HaloCaMP₅₈₅ reconstituted with JF₅₈₅-HTL exhibited a brightness of 90 mM⁻¹cm⁻¹, substantially higher than LAR-GECO1 (7.2 mM⁻¹cm⁻¹) and R-CEPIAer (6.2 mM⁻¹cm⁻¹) (Table 1), confirming a nearly 15-fold increase in molecular brightness relative to currently available red GECIs. We next expressed LAR-GECO, R-CEPIAer and ER-HaloCaMP₅₈₅ in HeLa cells and primary cortical neurons and using identical illumination and detection conditions we compared their relative *in-cell* brightness. We found that ER-HaloCaMP₅₈₅ was approximately 100-fold brighter in HeLa cells and 20-fold brighter in somata of primary neurons (Fig. 1F, G), supporting data from purified protein. Importantly, these differences in live cell imaging cannot be ascribed to our optical setup, as it is actually better matched to excite and collect fluorescence from LAR-GECO1 and R-CEPIAer (see Methods). This suggests that the enhanced detectability of ER-HaloCaMP₅₈₅ in cells is driven by its intrinsically higher brightness.

Next, we investigated the ability of these sensors to detect ER fluorescence within the single, narrow ER tubules of axons, a task that represents a significant challenge in ER Ca²⁺ imaging. This comparison provided a substantial improvement over previous technologies, as neither LAR-GECO nor R-CEPIAer enabled axonal ER detection in any neuron presenting somatic ER labeling, whereas ER-HaloCaMP₅₈₅ allowed us to identify it clearly in over 50% of the transfected neurons when using identical imaging conditions (Fig. 1G, H; see Methods). These results indicate that ER-HaloCaMP₅₈₅ provides a significant improvement in red ER Ca²⁺ imaging by providing significantly better responsiveness and brightness.

ER-HaloCaMP enables ER Ca²⁺ measurements in locales with low ER content

As ER-HaloCaMP₅₈₅ is the only red ER Ca²⁺ sensor visible in axons of neurons, we sought to evaluate its responsiveness in this small compartment. When neurons fire action potentials, axonal ER uptakes Ca²⁺ from the cytosol through sarcoplasmic/endoplasmic reticulum Ca²⁺-ATPases (SERCAs)^{10,12,56}. Using field stimulation (see Methods), we electrically stimulated primary cortical neurons to mimic physiological firing paradigms of neurons *in vivo*, such as firing at 20 Hz for 1s⁵⁷, and quantified axonal ER Ca²⁺ dynamics using ER-HaloCaMP₅₈₅. We observed robust activity-driven ER Ca²⁺ uptake, which was completely silenced when inhibiting SERCA pumps using thapsigargin, confirming the specificity of the measurements (Fig. 2A, B). Since other red ER Ca²⁺ sensors were not visible in axons, we compared the performance of ER-HaloCaMP₅₈₅ with the current gold standard ER Ca²⁺ sensor, the GFP-based ER-GCaMP6-210¹². ER-HaloCaMP₅₈₅ exhibited uptake responses comparable to those of ER-GCaMP6-210, demonstrating its high efficacy in measuring ER Ca²⁺ dynamics within small ER volumes (Fig. 2C, D).

To examine whether ER-HaloCaMP₅₈₅ could accurately detect physiological ER Ca²⁺ release (decrease in signal) and not only uptake (increase in signal), we assessed its responsiveness in neuronal dendrites, another subcellular localization containing limited ER volume. During excitatory neurotransmission, dendrites receive extracellular glutamate inputs from other neurons, which drive Ca²⁺ entry from the extracellular space and cause Ca²⁺ release from dendritic ER^{24,58,59}.

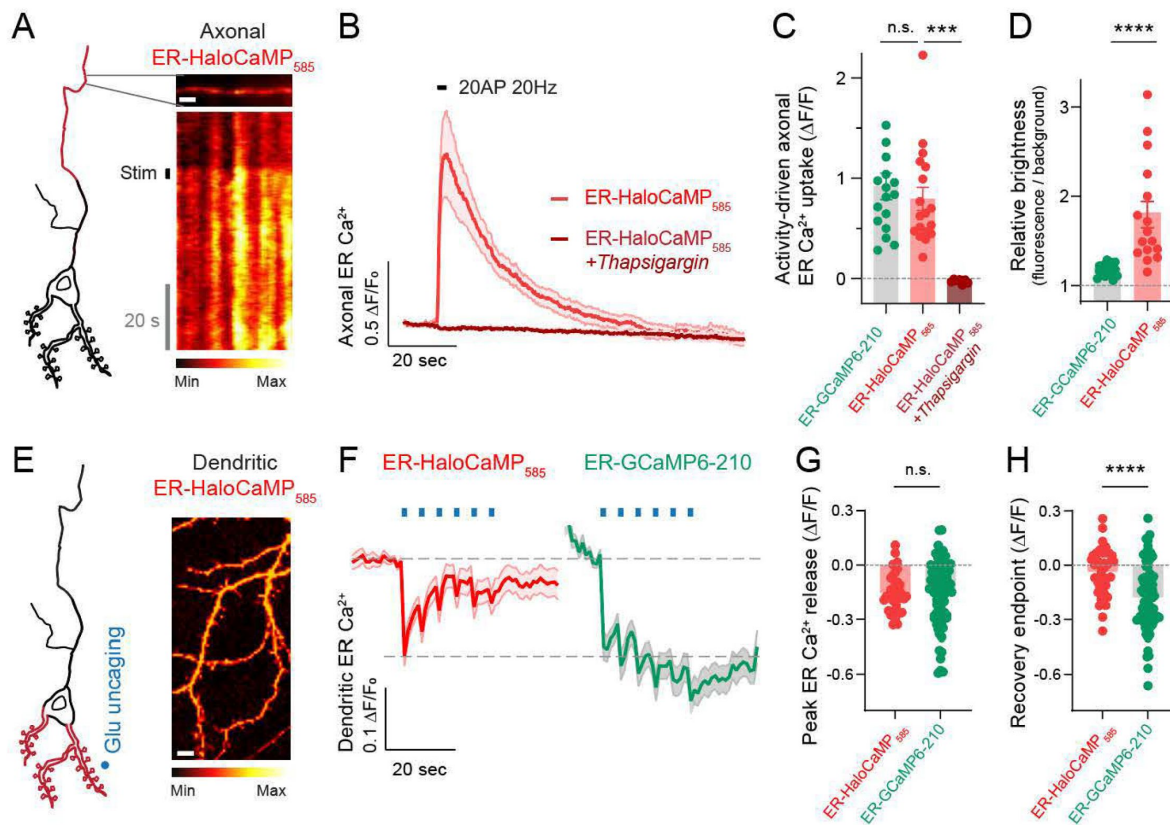


Figure 2.

ER-HaloCaMP₅₈₅ enables ER Ca²⁺ measurements in subcellular locales with low ER content.

(A-D) Axonal ER Ca²⁺ responses to stimulation of 20 action potentials (AP) at 20 Hz measured using different indicators. **(A)** On the left: schematic representation of ER-HaloCaMP₅₈₅ expressed within a single ER tubule of axon. Right panel shows a representative image of an axon expressing ER-HaloCaMP₅₈₅ and changes in fluorescence induced by stimulation. Scale bar, 4 μ m. Pseudocolor scale of intensity shown below. **(B)** Axonal ER Ca²⁺ responses to stimulation of 20AP at 20 Hz before and after 15 minute thapsigargin treatment using ER-HaloCaMP₅₈₅. **(C)** Quantification of activity-driven ER Ca²⁺ uptake peak response upon 20 AP (20 Hz) stimulation using ER-GCaMP6-210 (green), ER-HaloCaMP₅₈₅ (red) and ER-HaloCaMP₅₈₅ (dark red) in the presence of thapsigargin. **(D)** Quantification of relative brightness measured with ER-GCaMP6-210 or ER-HaloCaMP₅₈₅ in axons. **(E-H)** Dendritic ER Ca²⁺ responses to glutamate uncaging in spines. **(E)** On the left: schematic representation of ER-HaloCaMP₅₈₅ in dendrites and location of glutamate uncaging on a spine. Right panel shows the dendrites of a neuron expressing ER-HaloCaMP₅₈₅, (pseudocolor scale relative to the image showing low to high intensity). Scale bar, 4 μ m. **(F)** Dendritic ER Ca²⁺ responses adjacent to the site of glutamate uncaging measured with ER-HaloCaMP₅₈₅ or ER-GCaMP6-210. Blue ticks indicate the 6 uncaging pulses of 100 ms at 0.25 Hz, 7.2 mW 720 nm uncaging pulses. **(G)** Quantification of dendritic ER Ca²⁺ peak release. **(H)** Corresponding quantification of fluorescence recovery 15 s after glutamate uncaging.

To study this process, we locally released glutamate near single dendritic spines of neurons in culture using glutamate uncaging^{60,61} and examined ER Ca^{2+} responses using ER-HaloCaMP₅₈₅. Our experiments revealed that ER-HaloCaMP₅₈₅ responds to single-glutamate uncaging pulses with distinct ER Ca^{2+} release events and a high signal-to-noise ratio. The relative magnitude of these events was indistinguishable from those obtained with ER-GCaMP6-210 (**Fig. 2E-G**, see Methods).

However, when comparing the photostability of prolonged measurements of ER Ca^{2+} release, we found that signals of ER-HaloCaMP₅₈₅ robustly recovered to the initial baseline, while ER-GCaMP6-210 showed a loss of fluorescence at the end of the experiment, suggesting reduced photostability (**Fig. 2H**). We also observed that ER-GCaMP6-210 exhibited a rapid decrease in fluorescence immediately after 488 nm light exposure both in dendrites and axons, which then quickly settled into a new steady-state baseline (**Fig. S2A, B**). While this phenomenon has been previously observed on ER-GCaMPs⁵⁸, its origin remains unclear. It may reflect a rapid photophysical transition, such as reversible photoswitching or photoisomerization, in which an initial exposure to blue light drives a fraction of the GFP-based molecules into a less fluorescent state, quickly reaching equilibrium⁶². Contrary to this behavior, ER-HaloCaMP₅₈₅ maintained a stable and bright baseline signal throughout the imaging period in both axons and dendrites, exhibiting superior photostability compared to ER-GCaMP6-210 (**Fig. S2A, B**).

ER-HaloCaMP fluorescence is tunable and enables far-red ER Ca^{2+} measurements

We next evaluated the usability of ER-HaloCaMP conjugated with JF₆₃₅-HTL (ER-HaloCaMP₆₃₅) for measuring ER Ca^{2+} levels in the far-red spectrum, which, to our knowledge, has not yet been possible. Conjugation with JF₆₃₅-HTL instead of JF₅₈₅-HTL led to excitation and emission spectra with a peak excitation at 643 nm and emission at 654 nm, confirming its usability as a far-red sensor (**Supp. Fig. S3A**). Using purified protein, LA-HaloCaMP₆₃₅ presented a shift in the Ca^{2+} affinity compared to LA-HaloCaMP₅₈₅, increasing it by approximately 2.5-fold (**Table 1**). ER-HaloCaMP₆₃₅ retained a molecular brightness of $48 \text{ mM}^{-1}\text{cm}^{-1}$ which remains substantially higher than that of conventional red GECIs (~ 7 -fold increase, **Table 1**). When imaged in HeLa cells, ER-HaloCaMP₆₃₅ generated a far-red fluorescence pattern consistent with ER labeling (**Fig. 3A**). Upon thapsigargin addition, fluorescence decreased by about 80%, confirming its usability as a far-red ER Ca^{2+} sensor (**Fig. 3B**). Given the non-existence of other far-red ER Ca^{2+} sensors, we evaluated ER-HaloCaMP₆₃₅ by comparing its $\Delta F/F$ changes to those of ER-GCaMP6-210 and ER-HaloCaMP₅₈₅. The results showed that the changes detected by ER-HaloCaMP₆₃₅ were approximately in the same range as those observed with ER-GCaMP6-210 or ER-HaloCaMP₅₈₅ (**Fig. 3C**). This result supports the versatility of ER-HaloCaMP in capturing live ER Ca^{2+} dynamics in different fluorescent colors.

We then investigated whether ER-HaloCaMP₆₃₅ could be used to visualize and monitor ER Ca^{2+} dynamics fluorescence within the narrow ER tubules of axons. We first confirmed that the high brightness of ER-HaloCaMP₆₃₅ enabled successful detection of axonal ER at rest (**Fig. 3D**). Next, we electrically stimulated neurons and observed robust axonal ER Ca^{2+} uptake. As control, signals were completely silenced by thapsigargin, confirming that the increases in fluorescence depended on ER SERCA pumps (**Fig. 3E**). Benchmarking against ER-GCaMP6-210 and ER-HaloCaMP₅₈₅ showed that ER-HaloCaMP₆₃₅ exhibited lower responses, which could be a consequence of its higher affinity (**Fig. 3F**). Despite this caveat, ER Ca^{2+} uptake was consistently detected in response to electrical stimulation suggesting that resting ER Ca^{2+} concentration is low enough in axons for this indicator to be useful for detecting changes in subcellular regions with low ER abundance.

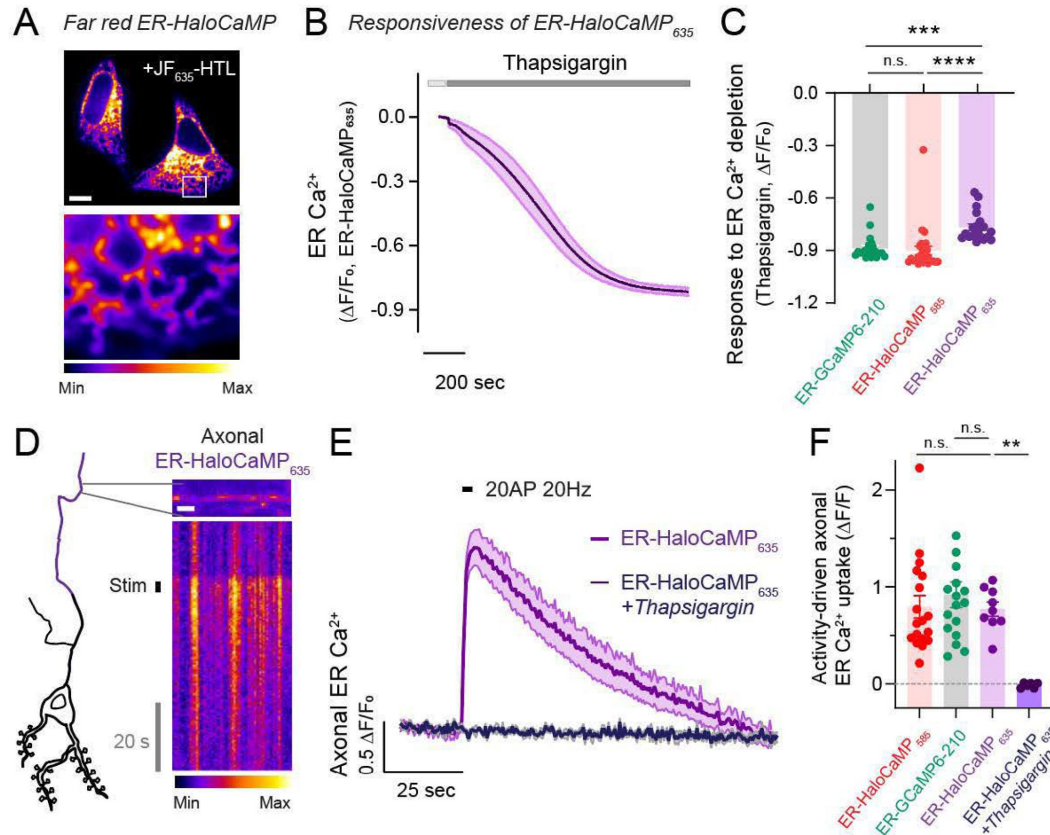


Figure 3.

Far-red ER Ca²⁺ measurements using ER-HaloCaMP₆₃₅.

(A) High resolution image of a HeLa cell expressing ER-HaloCaMP635 reconstituted with JF₆₃₅-HTL shows the ER structure (pseudocolor scale below showing low to high intensity). Scale bar, 12 μ m. (B) Average fluorescence intensity over time, upon thapsigargin treatment (1 μ M) in HeLa cells, measured with ER-HaloCaMP₆₃₅. Light gray indicates cells in 'fryode's solution, dark gray indicates when thapsigargin is added. (C) Quantification of the thapsigargin response (1 μ M) in HeLa cells, measured with different sensors. (D-F) Axonal ER Ca²⁺ responses to 20 AP (20 Hz) stimulation, using different indicators. (D) On the left: schematic representation of ER-HaloCaMP₆₃₅ expressed within a single ER tubule of axon. On the right representative image of a neuron expressing ER-HaloCaMP₆₃₅, zoomed on an axon. Scale bar, 4 μ m. Corresponding kymograph (pseudocolor scale below showing low to high intensity). (E) Corresponding fluorescence intensity over time, before and after thapsigargin treatment (1 μ M). (F) Quantification of activity-driven ER Ca²⁺ uptake peak response upon 20 AP (20 Hz) stimulation. Data are represented as mean $\pm$ SEM. See Supplementary Table STI for details on statistical tests and sample sizes.

Developing a responsive and tunable red/far-red mitochondrial Ca^{2+} sensor based on BaloCaMP

Building on the robust performance of ER-HaloCaMP, we extended our strategy to develop mitochondrial Ca^{2+} sensors based on HaloCaMPs. Mitochondrial Ca^{2+} entry occurs dynamically in cells and is essential for cellular physiology, as it adjusts energy production, modulates metabolic pathways and influences cell survival and apoptosis¹⁸. Resting free Ca^{2+} levels in mitochondria are higher than in the cytosol, yet remain in the hundreds of nanomolar rather than micromolar range^{11,63-65}. Previous studies have shown that low-affinity mitochondrial Ca^{2+} sensors exhibit inadequate responses to capture physiological stimuli in HeLa cells¹⁹ and neurons⁶⁶. In contrast, adapted sensors originally developed for cytosolic Ca^{2+} measurements give robust mitochondrial responses^{11,66}. We targeted HaloCaMP₄₀ to mitochondria using tandem mitochondrial targeting sequences from cyclooxygenase 8¹¹ and fused it with mTagBFP2 to create Mito-HaloCaMP (**Fig. 4A**). We expressed Mito-HaloCaMP in HeLa cells and conjugated it with JF₅₈₅-HTL, which generated a bright mitochondrial pattern in HeLa cells that upon histamine addition reported a robust change in mitochondrial Ca^{2+} (**Fig. 4B**). To assess the efficacy of Mito-HaloCaMP₅₈₅, we compared it against Mito-jRCaMP1b^{11,67} by measuring their respective $\Delta F/F_0$ responses and brightness under identical conditions. Our results demonstrated that Mito-HaloCaMP₅₈₅ not only exhibited significantly larger $\Delta F/F_0$ changes but was also significantly brighter, indicating enhanced sensitivity and improved signal-to-noise for monitoring mitochondrial Ca^{2+} dynamics in HeLa cells (**Fig. 4B-D**).

In non-excitable cells and neuronal somata, mitochondria are easily detectable and form networks of elongated structures. However, in axons their morphology is significantly reduced and they appear as small rounded structures averaging approximately 1 μm in length, thereby making their detection significantly more challenging. We next used Mito-HaloCaMP₅₈₅ to identify axonal mitochondria and electrically stimulated neurons to fire 20AP at 20 Hz as before (**Fig. 4E**). During neuronal firing, we observed that Mito-HaloCaMP₅₈₅ exhibited robust mitochondrial Ca^{2+} responses, generating fluorescence changes that were 3.5-fold greater than those obtained with Mito-jRCaMP1b under identical conditions (**Fig. 4F**). Similarly, when stimulating neurons with 100AP fired at 100 Hz, Mito-HaloCaMP₅₈₅ still reported significantly larger changes (**Fig. 4F,G**). These findings indicate that Mito-HaloCaMP₅₈₅ represents a significant advancement in mitochondrial Ca^{2+} imaging technology, providing a brighter and more responsive tool for red mitochondrial Ca^{2+} measurements.

We next evaluated the usability of Mito-HaloCaMP conjugated with JF₆₃₅-HTL (Mito-HaloCaMP₆₃₅) for measuring mitochondrial Ca^{2+} dynamics in the far-red spectrum, which, to our knowledge, has not yet been possible. We reasoned that the ability of measuring organellar Ca^{2+} handling in the far-red spectrum should enable quantifying Ca^{2+} dynamics in the different subcellular compartments involved in ER-mitochondria Ca^{2+} transfer, enabling the simultaneous detection of Ca^{2+} in three independent subcellular locales. We used histamine as a stimulus to drive ER Ca^{2+} release and simultaneously quantified Ca^{2+} dynamics in ER, cytosol and mitochondria of HeLa cells (**Fig. 5A**). These experiments were performed in the absence of extracellular Ca^{2+} to ensure that all observed signals originate from the ER, thereby allowing precise quantification of how much ER Ca^{2+} is released into the cytosol versus taken up by mitochondria (see Methods). We examined the timing of Ca^{2+} changes in each compartment using the cytosolic peak as a reference point and we found that ER Ca^{2+} release occurred earlier, while mitochondrial Ca^{2+} signals lagged several seconds behind (**Fig. 5B, C**). These results confirm that Ca^{2+} release from the ER is the initiating event that drives subsequent cytosolic and mitochondrial Ca^{2+} accumulation. Single-cell measurements revealed that the amplitude of both cytosolic and ER Ca^{2+} correlated with the amplitude of mitochondrial Ca^{2+} increases (**Fig. 5D**; **Fig. S3B, C**). However, we observed significant variability, finding large differences in ER-mitochondrial Ca^{2+} transfer in between different cells (**Fig. 5E**, top panel; **Fig. S3A, B**). As a control, we confirmed that differences did

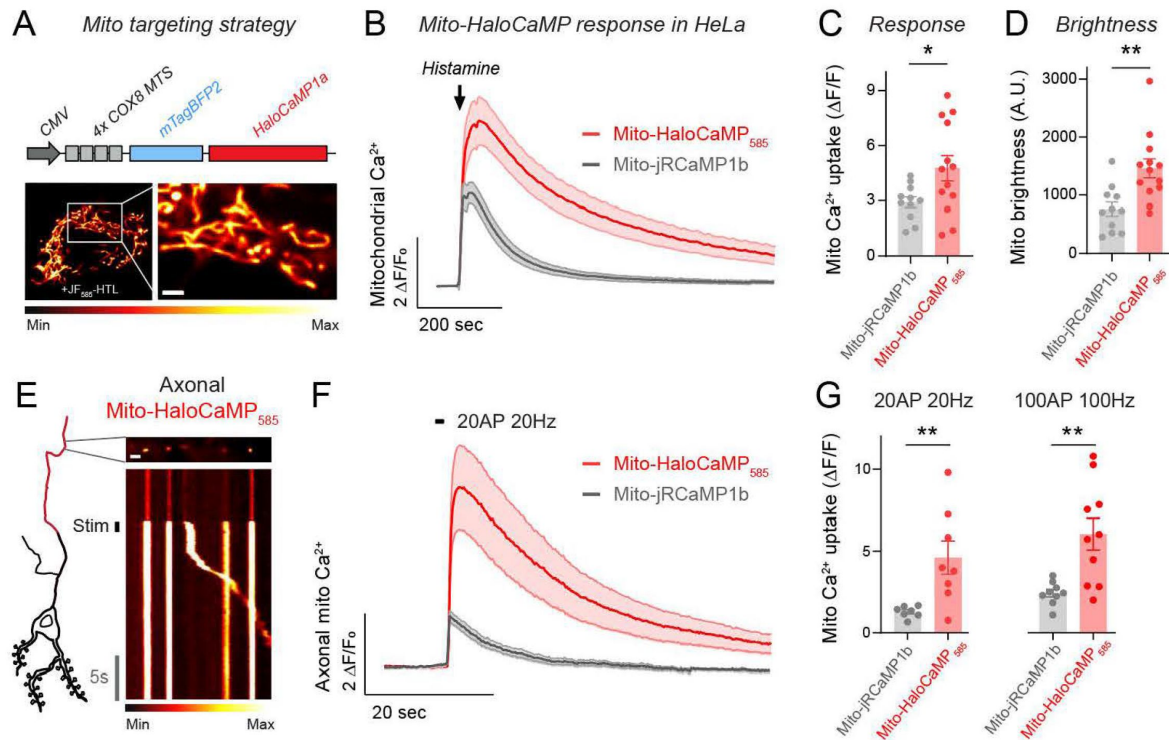


Figure 4.

Mito-HaloCaMP is a tunable and highly responsive mitochondrial Ca^{2+} sensor.

Top: targeting scheme for expression in the mitochondria by adding 4 times COX8 mitochondrial targeting sequence (MTS). Bottom: high resolution image of a HeLa cell expressing Mito-HaloCaMP reconstituted with JF₅₈₅-HTL shows the mitochondrial structures (pseudocolor scale below showing low to high intensity). Scale bar, 2 μm . **(B)** Average fluorescence intensity over time, upon histamine treatment (10 μM), measured with red mitochondrial calcium indicators in HeLa cells. **(C)** Corresponding quantification of the peak response. **(D)** Comparison of relative brightness of mito-RCaMP1b and mito-HaloCaMP₅₈₅ using identical illumination and detection conditions in intact HeLa cells. **(E)** On the left: schematic representation of mito-HaloCaMP₅₈₅ expressed within a single axon. On the right representative image of a neuron expressing mito-HaloCaMP₅₈₅, zoomed on an axon. Scale bar, 4 μm . Corresponding kymograph (pseudocolor scale below showing low to high intensity). **(F)** Corresponding average fluorescence intensity over time upon 20 AP (20 Hz) stimulation, measured with red mitochondrial calcium indicators in axons of cortical neurons. **(G)** Quantification of activity-driven mitochondrial Ca^{2+} uptake peak response upon 20 AP (20 Hz) stimulation or 100 AP (100 Hz) with red mitochondrial calcium indicators.

not arise from saturation of the sensors during histamine responses, as we used ionomycin at the end of every experiment to obtain the maximal response of each sensor (**Fig. S3D-G**). To quantify ER-mitochondrial Ca^{2+} coupling, we leveraged our simultaneous quantification of ER, mitochondria and cytosol to quantify in individual cells the distribution of ER-derived Ca^{2+} into mitochondrial and cytosolic compartments (Fig. SE, bottom; see Methods). Our results show that as ER Ca^{2+} release increases, cytosolic Ca^{2+} increases and the coupling to mitochondrial Ca^{2+} entry increases cooperatively. This indicates that ER-mitochondria Ca^{2+} transfer is not a linear process but it is dynamically adjusted with variable ER Ca^{2+} release events. This novel strategy to monitor ER-mitochondria Ca^{2+} transfer provides a robust technical framework to explore the fitness of this process in different cell types and cell states in health and disease.

Lastly, we examined the capabilities of Mito-HaloCaMP₆₃₅ to detect mitochondrial Ca^{2+} uptake in single mitochondria of the axon during neuronal activity (Fig. SF). We electrically stimulated neurons to fire 20AP at 20 Hz or 100AP at 100 Hz as before and observed that Mito-HaloCaMP₆₃₅ exhibited robust mitochondrial Ca^{2+} responses (**Fig. 5G, H**). As no far-red mitochondrial Ca^{2+} sensors currently exist, we compared Mito-HaloCaMP₆₃₅ to Mito-GCaMP6f, the gold standard in the field. We found that Mito-GCaMP6f responses were indistinguishable from those of Mito-HaloCaMP₆₃₅, demonstrating its high sensitivity (**Fig. 5I**). We also compared these data to Mito-HaloCaMP₅₈₅, which performed significantly better than Mito-GCaMP6f (**Fig. 5I**). These results show that Mito-HaloCaMP enables robust detection of mitochondrial Ca^{2+} changes in red and far-red even in challenging imaging conditions, such as detecting fluorescence changes in single small axonal mitochondria.

ER-HaloCaMP Ca^{2+} measurements in brain tissue of different species

Lastly, to extend the applicability of organellar HaloCaMPs beyond cultured cells, we next assessed the ability of ER-HaloCaMP₅₈₅ to report ER Ca^{2+} dynamics in *ex vivo* brain preparations from rats and *Drosophila*. We first expressed ER-HaloCaMP in individual CA3 pyramidal neurons in organotypic rat hippocampal slices (**Fig. 6A**). After labeling the sensor with JF₅₈₅-HTL, we triggered and quantified somatic ER Ca^{2+} release via DHPG (3,5-dihydroxyphenylglycine), a group I metabotropic glutamate receptor agonist that activates the production of inositol 1,4,5-trisphosphate (IP_3) and the release of Ca^{2+}

from internal ER stores^{68,69}. Brief application of DHPG led to a transient reduction in neuronal ER Ca^{2+} levels, showing that ER-HaloCaMP₅₈₅ can successfully report ER Ca^{2+} dynamics in thick samples using two-photon microscopy (**Fig. 6B**). Next, we evaluated ER-HaloCaMP₅₈₅ functionality in *Drosophila* neurons. Specifically, we targeted sensor expression to Myosuppressin (Ms) neurons within the *pars intercerebralis*, an evolutionarily conserved neuroendocrine center of the fly brain (**Fig. 6C**). Following labeling with JF₅₈₅-HTL, ER-HaloCaMP fluorescence was robustly detectable within the somata of Ms neurons (**Fig. 6D**). Treatment with thapsigargin resulted in a clear and progressive decrease in ER Ca^{2+} levels in Ms neurons as reported by the decrease of ER-HaloCaMP₅₈₅ fluorescence (**Fig. 6E**), while the fluorescence of mTagBFP2 remained unaltered (**Fig. S4A-C**). These results demonstrate that ER-HaloCaMP can report ER Ca^{2+} dynamics across tissues from evolutionarily distant species, validating its versatility as a tool for studying ER Ca^{2+} signaling in intact tissue.

Discussion

The interplay between cytosolic and organellar Ca^{2+} pools orchestrates a myriad of cellular responses essential for cellular function and survival^{2,70} and alterations in organellar Ca^{2+} signaling are heavily associated with disease^{18,71,75}. However, much of our understanding

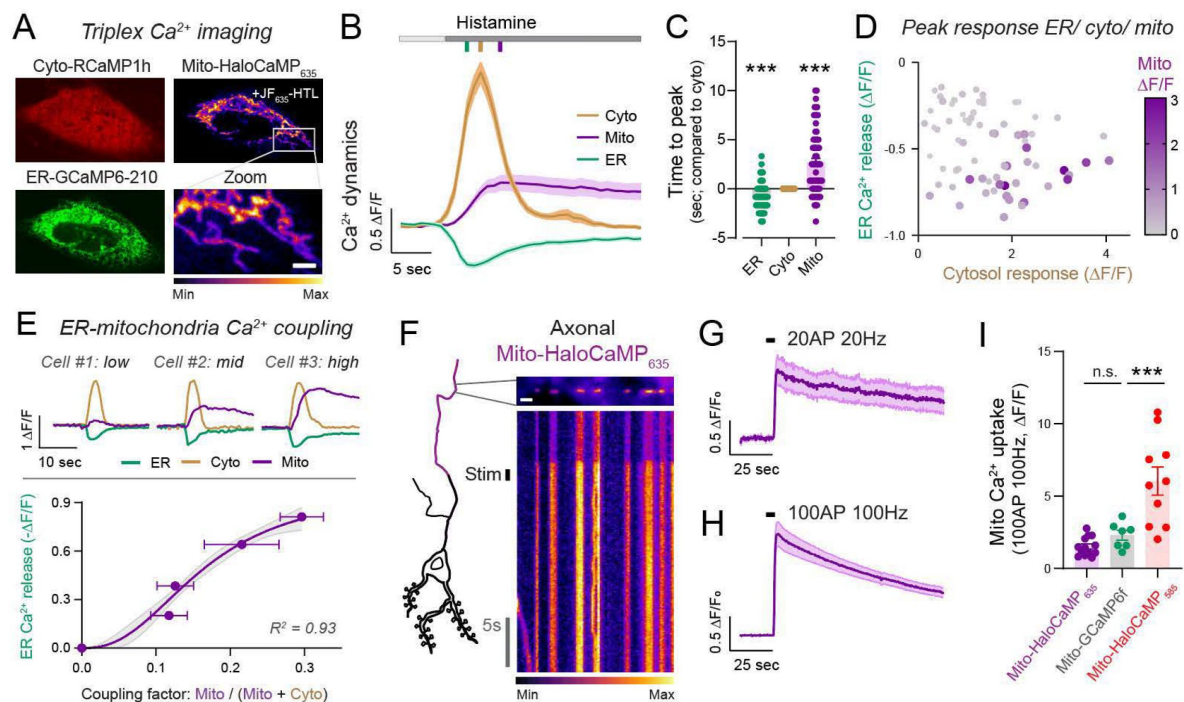


Figure 5.

Far-red mit(rHaloCaMP imaging enables multiplexing Ca^{2+} signaling.

(A) HeLa cell expressing cytosolic RCaMP1h (red), ER-GCaMP6-210 (green) and MitoHaloCaMP₆₃₅. Zoomed image shows individual mitochondria. Intensity bar: low to high fluorescence. Scale bar, 2 μm . **(B)** Averaged responses to histamine (dark grey) in the cytosol (cyto, brown), mitochondria (mito, purple), and endoplasmic reticulum (ER, green). Vertical colored bars indicate the time at which each organelle response is maximal. **(C)** Time-to-peak Ca^{2+} responses in each compartment, calculated relative to the cytosolic peak set to $t=0$. **(D)** Correlation between ER, cytosolic and mitochondrial Ca^{2+} signals during histamine. Each dot represents a single cell. Dot size and color intensity are scaled to reflect mitochondrial $\Delta F/F$. **(E)** Top panel: single cell examples showing low, medium and high mitochondrial Ca^{2+} responses for similar ER and cytosol Ca^{2+} dynamics. Lower panel: quantification of ER-mitochondrial Ca^{2+} transfer during histamine. Mitochondria and cytosol peak responses were used to calculate the coupling factor (see Methods). The line represents a fit to a Hill curve, and the gray shading indicates the 99% confidence interval of the fit. R^2 (coefficient of determination) = 0.93. **(F-1)** Axonal Mito-HaloCaMP₆₃₅ Ca^{2+} responses to 20 AP (20 Hz) and 100 AP (100 Hz) stimulation. **(F)** On the left: schematic representation of Mito-HaloCaMP₆₃₅ expressed within a single axon. On the right: representative image of a neuron expressing Mito-HaloCaMP₆₃₅, zoomed on an axon. Scale bar, 4 μm . Corresponding kymograph of response, scale bar indicates 5 sec. Pseudocolor scale shows low to high intensity. **(G)** Average fluorescence intensity over time upon 20 AP (20 Hz) stimulation. **(H)** Average fluorescence intensity over time upon 100 AP (100 Hz) stimulation. **(I)** Quantification of activity-driven mitochondrial Ca^{2+} uptake upon 100 AP (100 Hz) measured with Mito-HaloCaMP₆₃₅, Mito-GCaMP6f and Mito-HaloCaMP₅₈₅. Data are represented as mean $\pm$ SEM. See Supplementary Table ST1 for details on statistical tests and sample sizes.

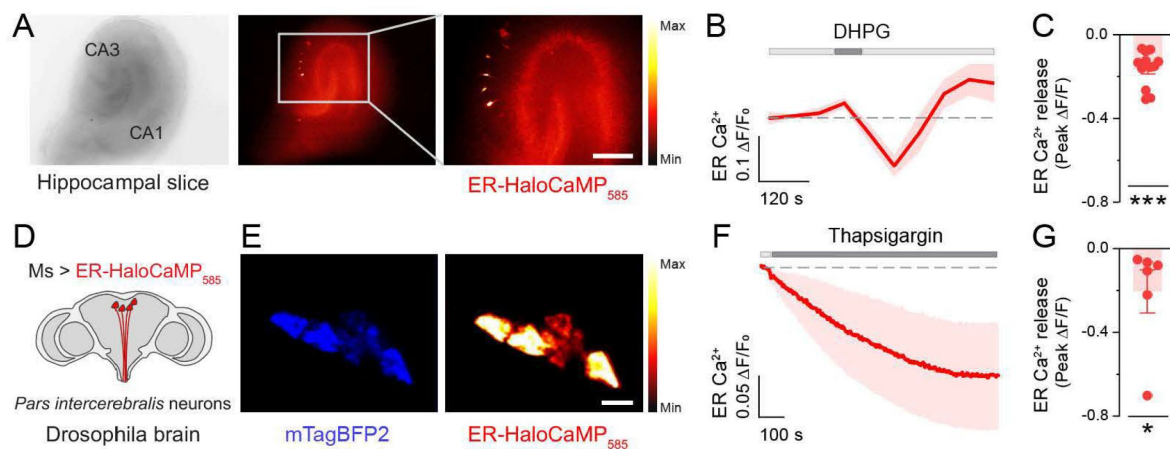


Figure 6.

Visualization of ER Ca^{2+} dynamics in rat and fly brain tissue using ER-HaloCaMP

(A) Representative images of an organotypic hippocampal slice with individual CA3 neurons expressing ER-HaloCaMP₅₈₅. A magnified view of neurons is shown on the right. Scale bar, 300 μm . **(B)** ER Ca^{2+} decreases after a pulse of 1 min application of 50 μM DHPG and overshoots later during washout. Line and shading are mean $\pm$ SEM. **(C)** Quantification of ER Ca^{2+} release by DHPG in individual neurons using ER-HaloCaMP₅₈₅. **(D)** Scheme representing the fly brain and the pars intercerebralis region in which Myosuppressin (Ms) neurons are located. ER-HaloCaMP is expressed in Ms neurons, denoted by Ms > ER-HaloCaMP₅₈₅. **(E)** Representative 2-photon images of mTagBFP2 and ER-HaloCaMP reconstituted with JF₅₈₅-HTL. **(F)** ER Ca^{2+} depletion by thapsigargin in Ms neurons using ER-HaloCaMP₅₈₅. **(G)** Quantification of ER Ca^{2+} depletion by thapsigargin in individual fly brains using ER-HaloCaMP₅₈₅. See Supplementary Table ST1 for details on statistical tests and sample sizes.

of organellar Ca^{2+} fluxes has relied on indirect approaches or on monitoring Ca^{2+} in different compartments in separate experiments due to limited multiplexing capabilities^{24,76,78}. For example, when studying ER-mitochondria Ca^{2+} transfer, dual-compartment Ca^{2+} imaging comparing cytosol and mitochondria, or cytosol and the ER, have been used to infer the relationship between cytosolic, ER and mitochondrial Ca^{2+} ,^{11,24,77} While these are highly valuable approaches, our triplex ER-cytosol-mitochondria Ca^{2+} measurements showed there is large cell-to-cell variability, suggesting that quantifying ER-mitochondria Ca^{2+} transfer in different populations of cells using only 2 sensors may complicate interpretations.^{15,18,79,80} The ability to image ER and mitochondrial Ca^{2+} using the far-red part of the spectrum unlocks distinguishing the relationship between the three compartments in single cells, enabling a precise quantification of ER-mitochondria Ca^{2+} transfer that takes into account the amplitude of cytosolic signals. Triplex imaging also enables the dissection of the temporal sequence of Ca^{2+} fluxes, revealing that during histamine stimulation, Ca^{2+} is first released from the ER, subsequently peaks in the cytosol, and is then taken up by mitochondria. Our experiments also reveal that ER Ca^{2+} release favors ER-mitochondria Ca^{2+} coupling in a cooperative manner. These findings in intact cells support the current model established in isolated mitochondria and permeabilized cells, in which opening of the Mitochondrial Ca^{2+} Uniporter (MCU) is controlled by cooperative Ca^{2+} binding to accessory Mitochondrial Calcium Uptake (MICU) proteins^{81,82} Overall, these experiments reveal the importance of simultaneous single-cell measurements of different compartments when studying organellar Ca^{2+} signaling and reveal the great potential of far-red imaging using organellar HaloCaMPs to explore complicated biological questions.

In addition to expanded multiplexing capacities, the development of ER-HaloCaMP and Mito-HaloCaMP presents 2 important advancements. First, they show significantly improved brightness and responsiveness in both ER and mitochondrial Ca^{2+} measurements in the red spectrum: the improved brightness of our organellar sensors likely reflects the higher extinction coefficients and quantum yields of synthetic dyes^{40,41}, as was shown when cytosolic HaloCaMP1₃₅ was compared to other protein-based fluorescent Ca^{2+} sensors, such as GCaMP6s⁴⁰. Enhanced brightness is of particular importance when measuring Ca^{2+} in organelles because quantifying Ca^{2+} dynamics in individual mitochondria^{11,13,14} or in narrow ER tubules¹² is a demanding task that only a few sensors are able to achieve. Second, they present improved photostability compared to the current best-in-class sensors: we found that upon blue light illumination, ER-GCaMPs underwent a rapid decrease in fluorescence that reaches stability in a few seconds, as shown previously^{5,8}. Such photoinactivation behavior has already been reported in some cytosolic GCaMPs, including preliminary versions of the original GCaMP, as well as GCaMP7c and jGCaMP8 variants^{8,3-8s}. Recent work on jGCaMP8 proposed that such behavior is the consequence of reversible photoswitching, likely caused by cis-trans isomerization of the GFP chromophore, which shifts the sensor to a non-fluorescent state independent of Ca^{2+} binding⁸³. This indicates that ER-GCaMPs likely suffer from the same problem, which distorts signals independently of ER Ca^{2+} changes and confounds the quantification of rapid Ca^{2+} transients at the start of the imaging period. Because ER-HaloCaMP uses an external synthetic fluorophore rather than a circularly permuted GFP, it cannot undergo the cis-trans isomerization that promotes photoswitching. Thus, ER-HaloCaMP shows significantly more stable fluorescence signals than ER-GCaMP while retaining similar Ca^{2+} responsiveness, making it the ideal choice for endoplasmic reticulum Ca^{2+} imaging.

Organellar HaloCaMPs also demonstrate robustness across diverse biological models (cultured cell lines, primary neurons and intact tissues from different species) and are compatible with both one- and two-photon excitation. This broad compatibility across model and imaging systems underscores the versatility of the HaloCaMP toolkit for investigating organellar Ca^{2+} signaling in diverse biological contexts and experimental conditions, which will facilitate their use in future research. Overall, we show that ER-HaloCaMP and Mito-HaloCaMP present improved properties that make them the preferred choice in several Ca^{2+} signaling paradigms in cell lines and primary

cells. Thus, our expanded palette of organellar Ca^{2+} sensors unlocks new possibilities for the study of organellar Ca^{2+} signaling across fields, which will facilitate our understanding of organellar cell physiology in health and disease.

Material and methods

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Jaime de Juan-Sanz (jaime.dejuansanz@icm-institute.org).

Molecular biology for generating the low affinity HaloCaMPs

Synthetic DNA oligonucleotides were purchased from Integrated DNA technologies. Q5 high fidelity DNA polymerase (New England Biolabs) was used for all PCR amplifications. Isothermal assembly reactions were performed with a NEBuilder HiFi kit (New England Biolabs). Small scale DNA isolation was performed with QIAprep Spin Miniprep Kit (Qiagen). The pRSET vector backbone was acquired from Life Technologies. Inserts and vector backbones were amplified by PCR amplification. Vector backbones and inserts were assembled by isothermal assembly with 10-30 base pair overlap, and sequence verified by Sanger sequencing (Azenta Life Sciences) or by nanopore full-plasmid sequencing (Plasmidsaurus).

Protein expression and purification

For expression and purification of proteins, T7 express (New England Biolabs) were transformed with pRSET plasmids encoding for the protein of interest. The bacteria were grown in auto-induction media using the Studier method⁸⁷ with antibiotics at 30 °C for 48 h shaking at 200 rotations per minute (r.p.m.). Cell pellets were collected by centrifugation, lysed in TRIS Buffered Saline (TBS; 19.98 mM Tris, 136 mM NaCl, pH 8.0), with n-Octyl- β -D-thioglucoopyranoside (1%). Aggregations were disrupted by sonication and the lysate cleared by centrifugation. Protein purification was performed on a Ni-nitrilotriacetic acid (NTA) resin using HisPur Ni-NTA resin (ThermoFisher Scientific), according to manufacturer's recommendation. Purified proteins were buffer exchanged into TBS using Amicon concentration filters (Merck). Protein aliquots were stored at 4 °C.

Calcium titrations in purified protein

To determine the calcium affinity and cooperativity of the low affinity HaloCaMP sensor, calcium titrations were performed in a buffer system made from CaNTA and NTA prepared using the pH titration method described by Tsien and Pozza⁸⁸ mixed in specific ratios to generate known free calcium concentrations. The free calcium concentration was calculated assuming the dissociation constant of NTA for Ca^{2+} 67 μ M. HaloCaMP proteins were pre-labeled with a limiting dye-HaloTag ligand (20 pM protein, 10 μ M JF585-HTL). 2 μ L of the pre-labeled protein-dye conjugate was diluted into 98 μ L of pre-mixed solutions of Ca-NTA in black 96-well plates. Fluorescence intensities were read on a plate reader (Tecan Spark 20M). Fluorescence intensity was measured at 26 °C and 37 °C. For HaloCaMPs with JF585-HTL excitation was 590 nm and emission 620 nm. For GCaMPs excitation was 484 nm and emission was 510 nm. All bandwidths were set to 5 nm. Changes in fluorescence in addition of Ca^{2+} were calculated in Microsoft Excel. The fluorescence (y) was plotted against the free calcium concentration (x) and a four-parameter dose-response curve (variable slope) using GraphPad Prism v10 software was fit where (a) is the value of fluorescence

Reagent or Resource	Source	Identifier
Experimental models: Organisms/Strains		
HeLa cells	LGC standards	ATCC-CRM-CCL-2
HEK 293T/17	LGC standards	ATCC-CRL-11268
Sprague-Dawley Rat	Charles River	Strain: Crl:CD(SD)
Ms-Gal4 <i>D. melanogaster</i>	Park et al., 2008 ⁸⁶	N/A
UAS-ER-HaloCaMP <i>D. melanogaster</i>	This paper	N/A
DNA constructs		
CMV Mito-HaloCaMP	This paper	Addgene plasmid #232195
CMV ER-HaloCaMP	This paper	Addgene plasmid #232198
CamKII Mito-HaloCaMP	This paper	Addgene plasmid #232196
CamKII ER-HaloCaMP	This paper	Addgene plasmid #232197
R-CEPIAer	Suzuki et al, 2014 ¹⁹	Addgene plasmid #58216
LAR-GECO1	Wu et al, 2014 ²¹	Addgene plasmid #61244
Mito4x-jRCaMP1b	Ashrafi et al, 2020 ¹¹	Addgene plasmid #127873
RCaMP1h	Hirabayashi et al, 2017 ²⁴	Addgene plasmid #105014

Resources Table.

at the bottom of the curve, (b) is the value of fluorescence at the top of the curve, (EC_{50}) is the concentration of agonist that gives a response halfway between bottom and the top, and (h) is the Hill cooperative coefficient.

$$y(x) = a + \frac{x^h \times (b - a)}{x^h + EC_{50}^h}$$

Sequence of LA-HaloCaMPI

(M)ARRKWQKTGHAVRAIGRLSSPEFAAETQAFRTTIDVGRKLIDQNVFIEGTLPMGVVRPLTEV
EMDHYREPFLNPVDREPLWRFNLPPIAGEPANIVALVEEYMDWLHQSPVKLLFWGTPGVLPF

AEAARLAKSLPNCKAVDIGPGLNLLQEDNPDIGSEIARWLSTLEISGGGTGGSGGTGGSGGTGG
SMAEIGTGFFPDPHYVEVLGERMHYVDVGPRDGTPLFLHGNPTSSYVWRNIIPHVAPTHRCIAP
DLIGMGKSDKPDLYFFDDHVRFMDAFIEALGLEEVVLVIHDWGSALGFHWAKRNPVRVKGIAP
MEFIRPIPTWDEWREFARDQLTEEQIAEFKEEFSLFDKGGDGTITTKELGTVMRSLGQNPTEAELQ
DMINEVDAGGDDGTIDFPEFLTMMARKMKYRDTEEEIREAFGVFDKDGNGYISAAELRHVMTNL
GEKLTDEEVDEMIREADIDGDGQVNYEEFVQMMTAK

Color legend: MLCK Peptide, cpHaloTag, Linker Calmodulin with low affinity mutations highlighted

Animals

The rats of either sex used in this study to prepare primary cultures were of the Sprague-Dawley strain, CrI (SD), bred by Charles River Laboratories following the international genetic standard protocol (IGS). All experiments conducted at the Paris Brain Institute strictly followed the guidelines set by the European Directive 2010/63/EU and the French Decree n° 2013-118 for the protection of animals used in scientific research. In the case of organotypic slices, rats were housed and bred at the University Medical Center Hamburg-Eppendorf (UKE). All procedures complied with German animal welfare regulations (Tierschutzgesetz der Bundesrepublik: Deutschland, TierSchG) and conformed to Directive 2010/63/EU. Experimental protocols were approved by the Behörde für Justiz und Verbraucherschutz (BN), Lebensmittelsicherheit und Veterinärwesen, Hamburg.

Flies were raised on a standard cornmeal/agar diet (6.65% cornmeal, 7.15% dextrose, 5% yeast, 0.66% agar, supplemented with 2.2% nipagin and 3.4mL/L propionic acid). All experimental flies were kept in incubators at 70% humidity and on a 12h light/dark cycle, at 25°C. Flies were transferred to fresh vials every 2 days, and fly density was kept to a maximum of 20 flies per vial.

Generation of UAS-ER-HaLo-CaMP::BFP transgenic line

The ER-HaloCaMP fragment was PCR amplified using the primers (5'-aagatcctctagaggtacccTTAGAGTTCATCCTTGCC-3') and (5'-actctgaatagggaattgggATGGGACTGTTGTCTGTTC-3'). The fragment was then cloned in the open pUASTattB vector through site-directed mutagenesis using the kit Q5-site directed mutagenesis (NEB E0554S). The pUASTattB vector was opened with restriction digestion using the restriction enzymes EcoRI-HF (NEB R3101S) and XhoI (NEB R0146S). The resulting pUASTattB-ER-HaloCaMP plasmid was used to establish transgenic lines through cpC-31 integrase mediated recombination (BestGene Inc.). The landing attP site used was VK37 PBac{yellow[+]-attP-3B}VK00037, BDSC Stock 9752.

Primary rat co-culture of postnatal cortical neurons and astrocytes

Primary co-cultures of cortical neurons and astrocytes were obtained as previously described with small modifications⁸⁹. P0 to P2 rats of mixed gender were sacrificed, and their brains were dissected in a cold HBSS-FBS solution (IX HBSS + 20% Fetal Bovine Serum) to isolate the cerebral

cortexes. These were then cut into small pieces for digestion and dissociation. The tissue was washed twice with 1X HBSS-FBS and 1X HBSS before being incubated in a trypsin-based digestion solution containing DNase I (Merck, D5025) for 15 minutes at room temperature. Trypsin (Merck, T1005) was neutralized with HBSS-FBS solution, followed by two washes with 1X HBSS-FBS and two with 1X HBSS. The tissue was then transferred to a dissociation solution (1X HBSS, 5.85mM MgSO_4) and dissociated into single cells by gentle pipetting. The cells were centrifuged at 13,000 rpm for 10 minutes at 4°C, and the pellet was resuspended in 1X HBSS solution. After a second centrifugation, the pellet was resuspended in a homemade warmed plating media consisting of MEM (Thermo Fisher Scientific, 51200038) supplemented with 20 mM Glucose (Merck, 08270), 0.1 mg/ml transferrin (Merck, 616420), 1% GlutaMAX (Thermo Fisher Scientific, 35050061), 24 $\mu\text{g}/\text{ml}$ insulin (Merck, 16634), and 10% FBS (Thermo Fisher Scientific, 10082147) and 2% N-21 (Bio-technie, AR008). Finally, cells were plated in sterilized cloning cylinders (Merck, C7983; 38,000 cells per cylinder) attached to coverslips (Diameter= 25 mm, Warner Instruments, 640705) that had been pre-coated with 0.1 mg/ml poly-ornithine (Merck, P3655). After 3-4 days, the neuronal media was replaced with a homemade feeding media, similar in composition to the plating media but containing 5% FBS and 2 μM cytosine P-d-arabinofuranoside (Merck, C6645) to inhibit glial growth. The primary co-cultures of cortical neurons and astrocytes were maintained at 37°C in a humidified incubator with 95% air/5% CO_2 until the imaging experiments, which were performed from days in vitro (DIV)12 to DIV21.

For cultures used in glutamate uncaging experiments shown in **Figure 2E-H**, conditions were as follows: hippocampal regions were dissected in ACSF containing (in mM) 124 NaCl, 5 KCl, 1.3 $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 1.25 $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, 2 CaCl_2 , 26 NaHCO_3 , and 11 Glucose (stored at 4°C) and stored in hibernate E buffer (BrainBits LLC, stored at 4 °C). Dissected hippocampi were dissociated using Papain Dissociation System (Worthington Biochemical Corporation, stored at 4 °C) with a modified manufacturer's protocol. Briefly, hippocampi were digested in papain solution (20 units of papain per ml in 1 mM L-cysteine with 0.5 mM EDTA) supplemented with DNase I (final concentration 95 units per ml) and shook for 30-60 min at 37°C, 900 rpm. Digested tissue was triturated and set for 3 min, following which the supernatant devoid of tissue chunks was collected. The supernatant was centrifuged at 300 ref for 5 min and the pellet was resuspended in resuspension buffer (1 mg of ovomucoid inhibitor, 1 mg of albumin, and 95 units of DNase I per ml in EBSS). The cells were forced to pass through a discontinuous density gradient formed by the resuspension buffer and the Ovomucoid protease inhibitor (10 mg per ml) with bovine serum albumin (10 mg per ml) by centrifuging at 600 rpm for 6 min. The final cell pellet devoid of membrane fragments was resuspended in Neurobasal-A medium (Gibco, stored at 4 °C) supplemented with Glutamax (Gibco, stored at -20 °C) and B27 (Gibco, stored at -20 °C). Cells were plated on poly-D-lysine coated coverslips mounted on MatTek dishes at a density of 30000-50000 cells/ cm^2 . Cultures were maintained at 37 °C and 5% CO_2 with feeding every 3 days using the same medium until transfection. Transfections were performed 12 days after plating by magnetofection using Combimag (OZ biosciences, stored at 4 °C) and Lipofectamine 2000 (Invitrogen, stored at 4 °C) according to manufacturer's instructions.

Preparation of Organotypic Hippocampal Slices

Organotypic hippocampal slices were prepared from Wistar rats of both sexes at postnatal days 5-7. Following dissection, the hippocampi were sectioned into 400 μm slices using a tissue chopper and transferred onto a porous membrane (Millicell CM, Millipore) for culture. Slices were maintained at 37 °C in an atmosphere of 5% CO_2 in a culture medium consisting of 394 mL Minimal Essential Medium (Sigma M7278), 100 mL heat-inactivated donor horse serum (Sigma H1 138), 1 mM L-glutamine (Gibco 25030-024), 0.01 mg/mL insulin (Sigma 16634), 1.45 mL 5 M NaCl (Sigma S5150), 2 mM MgSO_4 (Fluka 63126), 1.44 mM CaCl_2 (Fluka 21114), 0.00125% ascorbic acid (Fluka 11140), and 13 mM D-glucose (Fluka 49152). The culture medium was partially replaced (60-70%) twice per week. To express ER-HaloCaMP in CA3 neurons of the cultured slice, plasmids encoding ER-HaloCaMP-BFP were diluted to a final concentration of 10 ng/ μL in a K-gluconate-based intracellular solution containing (in mM): 135 K-gluconate, 4 MgCl_2 , 4 N -ATP, 0.4 Na-OTF, 10

N -phosphocreatine, 3 ascorbate, and 10 HEPES (pH 7.2). Single-cell electroporation was performed between DIV 6 and DIV 10. During the electroporation procedure, slice cultures were maintained in a pre-warmed, HEPES-buffered solution containing (in mM): 145 NaCl, 10 HEPES, 25 D-glucose, 1 MgCl₂, and 2 CaCl₂ (pH 7.4, sterile filtered). Electroporation was carried out using an ELeCTroPORATOR (NPI), applying 12 voltage pulses (– 12 V, 0.7 ms) at 50 Hz. Slices were incubated for approx. 24h with JF₅₈₅-HTL in culture medium 5-7 days after electroporation, then rinsed in the pre-warmed HEPES-buffered solution before imaging. We found that ~40% of ER-HaloCaMP transfected neurons in the organotypic slices were not sufficiently labelled by incubation with JF₅₈₅-HTL, presenting detectable BFP while no ER-HaloCaMP₅₈₅ fluorescence. This could be the consequence of the limited bioavailability of JF₅₈₅-HTL⁴¹, although lack of labeling was not observed when reconstituting ER-HaloCaMP in *Drosophila* neurons.

Cell line culture and transfection

HeLa and HEK cells were cultured either on poly-ornithine coated glass coverslips or wells of a chambered coverslip with 8 wells and a glass bottom (CliniSciences, 80807-90). They were cultured in DMEM (Thermo Fisher Scientific, 10566016) supplemented with 10% fetal bovine serum and kept in an incubator with humidified air containing 5% CO₂ at a temperature of 37°C. HeLa and HEK cells were transfected using Lipofectamine 2000 (Invitrogen) the day before imaging following the manufacturer's recommendations.

HaloCaMP labelling in cultured cells

JF dye-ligands were obtained from Dr. Lavis (Janelia Research Campus; JF₅₈₅-HTL) or from Promega (HT 1050, Janelia Fluor 635 HaloTag Ligand; JF₆₃₅-HTL). The JF dye-ligands obtained from Promega were incubated following the manufacturers instructions. JF dye-ligands obtained from Dr. Lavis were received in aliquots of 100 nmol and resuspended in 200 µL DMSO to yield a 500 µM stock. JF dye-ligands were then diluted in the cell culture media (either DMEM for cell lines or feeding media for primary neuronal cultures or organotypic slices) at a final concentration of 1µM. The cells are usually labelled after 30 minutes of incubation but for practical purposes, they were incubated overnight with the dye-ligand. Next day, the dye-ligand was washed in the morning before the experiment. To minimize non-specific labeling and reduce background fluorescence, a series of washing steps were performed by replacing the existing media with fresh, pre-warmed culture media, followed by a 10-minute incubation period. This washing step was repeated twice to ensure thorough removal of non-specific signals and to facilitate subsequent fluorescence measurements.

Calcium imaging in cells and primary neurons

Live imaging assays of primary cortical neurons transfected with calcium phosphate method⁹¹ were conducted from DIV12 to DIV21. HEK or HeLa cells imaging assays were performed one day after transfection using lipofectamine 2000 (Thermo Fisher Scientific, 11668019). The experiments utilised a custom-built, laser-illuminated epifluorescence microscope (Zeiss Axio Observer 3) paired with an Andor iXon Life camera (model IXON-L-897), cooled to -70°C. Illumination was provided by fiber-coupled lasers at wavelengths of 488 and 561 nm (Coherent OBIS 488 nm and 561 nm), combined using the Coherent Galaxy Beam Combiner. Laser illumination was controlled by a custom Arduino-based circuit that synchronized imaging and illumination. Neuron-astrocyte co-cultures, HEK or HeLa on coverslips were placed in a closed bath imaging chamber for field stimulation (Warner Instruments, RC-21BRFS) and imaged with a 40x Zeiss oil objective Plan-Neofluar with a numerical aperture of 1.30 (WD = 0.21 mm). Neurons were stimulated using 1 ms current pulses between platinum-iridium electrodes in the stimulating chamber (Warner Instruments, PH-2), driven by a stimulus isolator (WPI, MODEL A382) controlled by the Arduino-based circuit. All the experiments were performed at 37°C. Temperature was kept constant using a Dual Channel Temperature Controller (Warner Instruments, TC-344C) that controlled the temperature of the stimulation chamber (Warner Instruments, PH-2) and simultaneously warmed

the imaging solutions using an in-line solution heater (Warner Instruments, SHM-6). Imaging solutions were flown at 0.35–0.40 mL/min. Imaging was performed using a Tyrode's solution composed of (in mM) 119 NaCl, 2.5 KCl, 2 CaCl₂, 2 MgCl₂, 20 Glucose together with 10 μ M CNQX and 50 μ M AP5, buffered to pH 7.4 at 37°C using 25 mM HEPES.

Multiplexed confocal calcium imaging in cell lines and analysis of ER-mitochondria coupling

HeLa cells expressing ER-GCaMP6-210, cytosolic RCaMP1h and Mito-HaloCaMP were cultured in chambered coverslips with 8 wells and a #1.5H glass bottom (Ibidi, 80827) and transfected using lipofectamine 2000 following manufacturer's instructions (Thermo Fisher Scientific, 11668019) to be imaged the next day. Live imaging assays were conducted with an inverted Leica spinning disk equipped with a Yokogawa CSU-X1 module and a Hamamatsu Orea Flash 4.0 sCMOS camera. The objective used was 63x, NA 1.4. The effective sampling rate at each wavelength was approximately 0.83 Hz, given a 1.2s interval between successive images. The samples were illuminated using lasers at 488 nm, 561 nm, and 637 nm. The emission was collected through 525/50, 607/35, and 685/40 filters mounted in an automated rotating filter wheel. Imaging was performed using a Tyrode's solution composed of (in mM) 119 NaCl, 2.5 KCl, 2 CaCl₂, 2 MgCl₂, 20 Glucose, buffered to pH 7.4 at 37°C using 25 mM HEPES.

Regions of interest (ROIs) were drawn over individual cells, covering ER, cytosol and mitochondria. Background-subtracted fluorescence time-courses ($\Delta F/F_0$) were calculated and peak responses were extracted from each channel individually after histamine addition. Cells typically showed multiple peak responses but only the first peak was used for analysis for consistency. A modified Tyrode's solution containing 4mM Ca²⁺ and 500 μ M ionomycin was added at the end of the experiment to confirm histamine responses did not saturate any of the sensors. When plotting triplex correlations, the size of the dots representing single cells ranged from 0.01 to 0.90 in arbitrary size units. Responses > 0.90 are shown at maximal dot size. ER-mitochondria coupling was quantified by calculating, for each cell initial response, the ratio of mitochondrial $\Delta F/F_0$ to the sum of mitochondrial plus cytosolic $\Delta F/F_0$. Data points were grouped into bins of 0.15 $\Delta F/F_0$ along the ER $\Delta F/F_0$ axis and the corresponding averaged coupling factors were plotted against ER $\Delta F/F_0$ to assess cooperativity. To obtain averaged responses, cytosolic, ER and mitochondrial $\Delta F/F_0$ traces were time-aligned by setting the histamine-induced cytosolic peak to 0, then averaged across cells. Only cells in which the peak was higher than six times the standard deviation of the baseline were included in this analysis (65 out of 71).

Measurement of dendritic ER-HaloCaMP and ER-GCaMP6-210 upon two-photon glutamate uncaging

For the two-photon glutamate uncaging experiments in **Figure 2E–H**, imaging was conducted 15–16 days after neuronal cell culture plating in a modified E4 imaging buffer containing: 120 mM NaCl, 3 mM KCl, 10 mM HEPES (buffered to pH 7.4), 4 mM CaCl₂, and 10 mM Glucose, lacking MgCl₂. Imaging was performed using a custom-built inverted spinning disk confocal microscope (3i imaging systems; model CSU-W1) attached to an Andor iXon Life 888. Image acquisition was controlled by SlideBook 6 software. Images were acquired with a Plan-Apochromat 63x/1.4 NA. Oil objective, M27 with DIC III prism, using a CSU-W1 Dichroic for 488/561 nm excitation with Quad emitter and individual emitters, at back aperture laser powers 2.00 mW (488 nm) and 2.65 mW (561 nm) for spine stimulation measurements. During imaging, the temperature was maintained at 37°C using an Okolab stage top incubator with temperature control.

For ER-HaloCaMP measurements, cytosolic GCaMP6s was co-transfected to confirm effective spine stimulation (data not shown). For ER-GCaMP6-210 measurement, neurons were co-transfected with PSD95-mCherry to identify spines for stimulation. Before glutamate uncaging, neurons were treated with 1 μ M TTX (Citrate salt, mM stock made in water, Abcam, ab1200552) and 2 mM 4-

Methoxy-7-nitroindolyl-caged-L-glutamate (MNI caged glutamate, Tocris Bioscience 1490, 100 mM stock made in modified E4 buffer) in modified E4 buffer lacking Mg^{2+} (see above). Glutamate uncaging was performed using a multiphoton-laser 720 nm (Mai TAI HP) and a Pocket cell (ConOptics) for controlling the uncaging pulses. Uncaging protocols of 6 uncaging pulses at 0.25 Hz with 100 ms pulse duration per pixel at 7.2 mW back aperture laser power was used.

To measure ER-HaloCaMP and ER-GCaMP6-210 response, $\sim 2 \mu m$ regions of interest (ROI) were drawn at the base of the stimulated spine. The average intensities were background subtracted using the intensity measured from an adjacent background area. For each successive time point during and after stimulation, the normalized intensity ($\Delta F/F$) was calculated using the equation: $\Delta F/F = (F - F_0)/F_0$, where F_0 is defined as the average fluorescence intensity of the consecutive 5 time points before spine stimulation, and F is defined as the fluorescence intensity measured at the time point of interest.

Ex vivo ER-HaloCaMP Ca^{2+} imaging in the *Drosophila* brain

Drosophila brain live imaging experiments were carried out as in Silva et al.⁹² with some modifications. Briefly, female flies carrying the Myosuppressin (Ms)-Gal4 driver⁹³ were crossed with male flies carrying the UAS-ER-HaloCaMP-BFP construct (which is fused to mTagBFP2, as in the case of experiments in cells in culture). Crosses for imaging experiments were raised at 25°C. 1- to 2-day-old adult progeny were used for each recording.

A single fly brain was dissected in haemolymph solution and mounted on a glass coverslip coated with poly-L-lysine (Sigma-Aldrich, P1524). The haemolymph solution contained 130mM NaCl (Sigma, S9625), 5 mM KCl (Sigma, P3911), 2 mM $MgCl_2$ (Sigma, M9272), 2 mM $CaCl_2$ (Sigma, C3881), 5 mM D-trehalose (Sigma, T9531), 30 mM sucrose (Sigma, S9378) and 5 mM HEPES-hemisodium salt (Sigma, H7637). Then, a total volume of 500 μL of haemolymph solution was added on top of the coverslip. Next, 1 μL of the JF58rHTL 500 μM stock solution was incubated for approximately 40 min prior imaging.

Two-photon imaging was performed on a Leica Stellaris 8 DIVE upright microscope equipped with a $\times 25$, 1.00-NA water-immersion objective. Two-photon excitation of mTagBFP2 and ER-HaloCaMP₅₈₅ was achieved using a Mai Tai eHP DeepSee MP laser tuned to 820 nm. Spectral separation was achieved by tuning detectors for mTagBFP2 (445-460nm) and JF585-HTL (580-650nm). Then, 512×512 images were acquired at 0.5Hz, and the entire duration of each recording was 1200 s. For Ca^{2+} depletion experiments, Thapsigargin (Sigma, T9033) was diluted in haemolymph solution to prepare a 100 μM stock solution. After 60 s of baseline acquisition, 5 μl of the Thapsigargin stock solution was added to the 500- μl bath solution on top of the brain, resulting in a final concentration of 1 μM .

Image analysis was performed using a custom-written MATLAB script⁹². Regions of interest (ROIs) were manually drawn around individual Myosuppressin neuronal somas. The average intensity values for mTagBFP2 and ER-HaloCaMP₅₈₅ channels over each ROI were calculated over time after background subtraction. The ratio was calculated by dividing ER-HaloCaMP₅₈₅ by mTagBFP2 intensity. Traces from all cell bodies were pooled for analysis.

Image analysis and statistics

We used the ImageJ plugin Time Series Analyzer V3 for imaging analysis of dynamic changes in fluorescence, except for ex vivo experiments in which we used a custom-written MATLAB script⁹². Statistical analysis was carried out using GraphPad Prism v10 for Windows, with specific tests indicated in the Supplementary Table ST1. For each dataset, normality was assessed using the Shapiro-Wilk test and then based on the result we chose either parametric or non-parametric tests for further comparisons. All n numbers, as well as the number of independent experiments, are detailed in the Supplementary Table ST1. Unless otherwise indicated, all data for

this study was acquired from at least 3 independent experiments. We used Python to calculate Spearman correlations and their associated p-values with the SciPy library. Pearson's correlation coefficients and associated p values were calculated using graphpad, and both were incorporated in a single heatmap using illustrator. For the figures, the mean and the standard error of mean are used. Results of statistical analysis are shown in figures corresponding to the following criteria: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, n.s., not statistically significant.

Supplementary figures

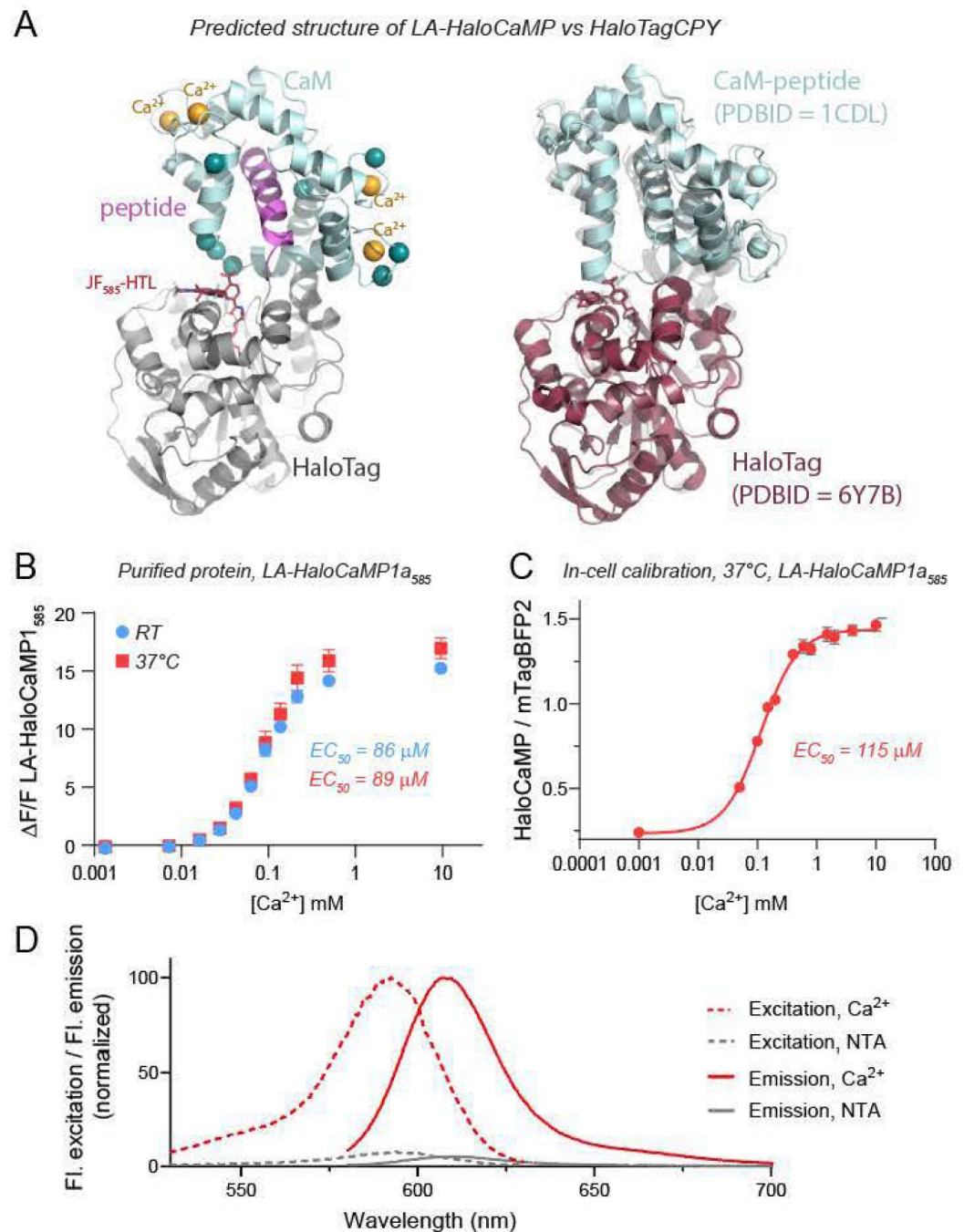


Fig. S1.

Biochemical properties of LA-HaloCaMPI.

(A) Predicted structural models of low-affinity HaloCaMP (LA-HaloCaMP) and HaloTagCPY generated using Chai Discovery. Left: LA-HaloCaMP is shown with HaloTag (gray), the CaM-binding peptide (purple), calmodulin (CaM, cyan) and bound calcium ions (yellow). JF₅₈₅,HTL is represented in red. Right: HaloTagCPY is modeled as a fusion of HaloTag (PDB ID: 6Y7B) and the CaM-peptide complex (PDB ID: 1CDL). (B) In vitro calcium titration of purified LA-HaloCaMP₅₈₅ protein at room temperature (RT, blue) and 37°C (red), showing EC₅₀ of 86 μM at room temperature (RT) and 89 μM at 37°C. (C) *In-cell* calibration at 37°C of LA-HaloCaMP₅₈₅ in permeabilized HeLa cells, yielding an EC₅₀ of 115 μM. (D) Excitation and emission spectra of LA-HaloCaMP labeled with JF₅₈₅,HTL in the presence of Ca²⁺ (red) or Ca²⁺-free buffered with NTA (gray). Excitation spectra (dashed lines) and emission spectra (solid lines) were normalized to their respective maxima. Data are represented as mean ± SEM. See Supplementary Table ST1 for details on sample sizes.

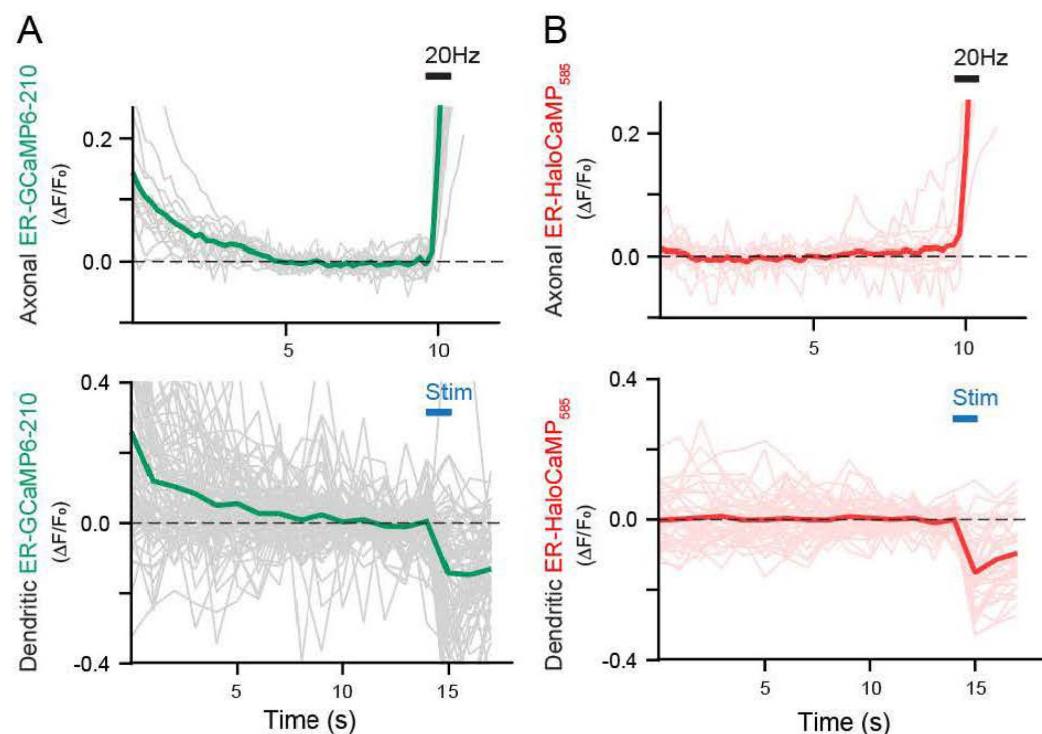


Fig. S2.

Photostability comparisons in axons and dendrites after light exposure

(A) Imaging of ER-GCaMP6-210 in either axons (top) or dendrites (bottom) shows a quick loss in fluorescence compatible with photoswitching of a fraction of the sensor population, resulting in a first decay in fluorescence that is quickly stabilized. (B) Comparative measurements of ER-HaloCaMP₅₈₅ in axons (top) or dendrites (bottom) show that signals were stable right after the illumination of the sample starts, showing improved photostability. Data are represented as mean ± SEM. See Supplementary Table ST1 for details on sample sizes.

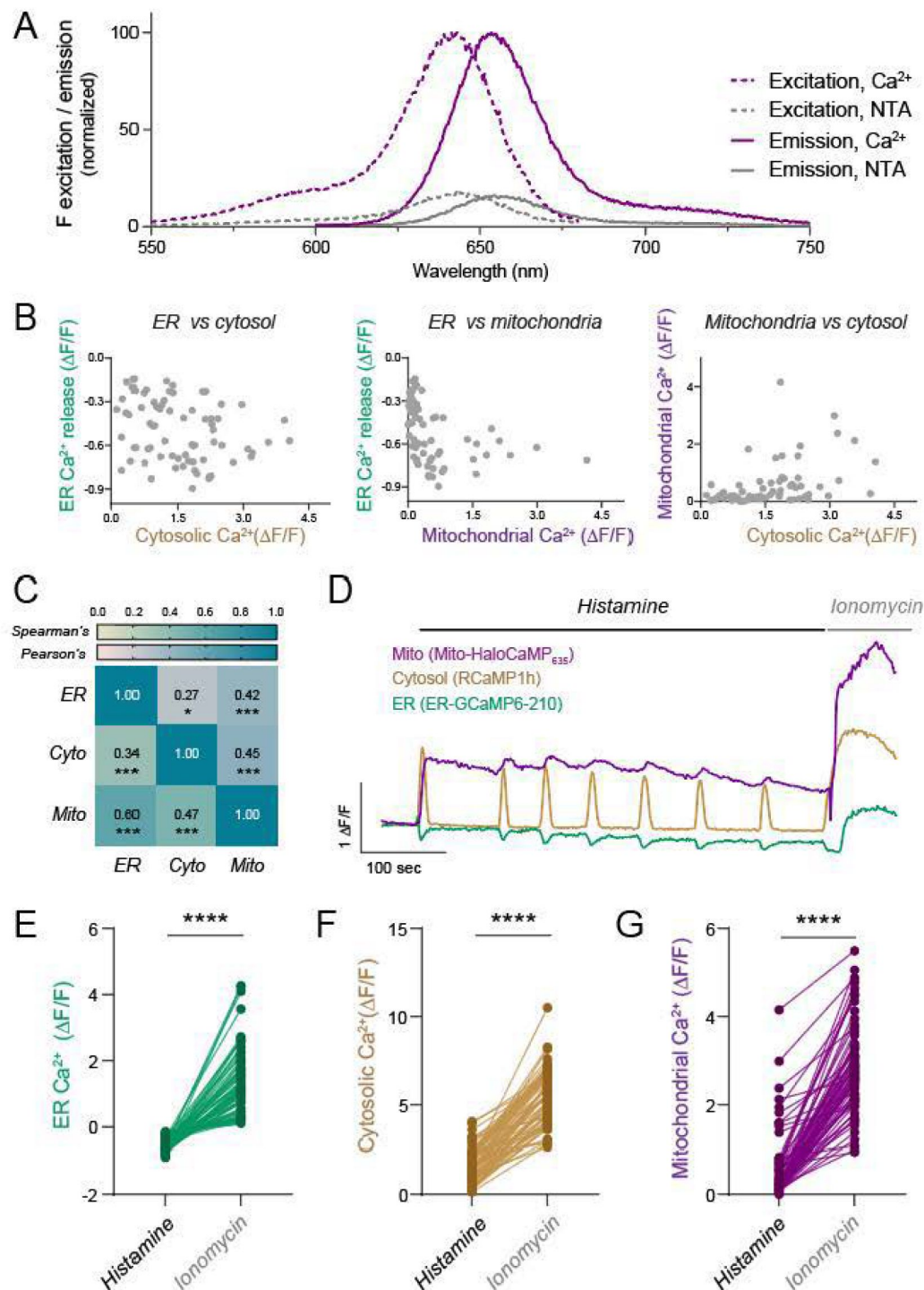


Fig. S3.

Correlation analysis of Ca^{2+} retpollIM and non-uturation of die sensors during histamine.

(A) Single-cell Ca^{2+} peak responses to histamine in theERand cytosol (left), ER and mitochondria (middle), or mitochondria and cy10S01 (right). (B) Correlation matrix of the data shown in (A), calculated under two assumptions: linear relationships (Pearson's) and mono10nic relationships (Spearman's). Each cell shows the respective correlation coefficient and asterisks indicate statistically significant deviation from zero. (C) Paired single-cell Ca^{2+}

peak responses in the ER (left), cytosol (middle), and mitochondria (right) after histamine and subsequent ionomycin treatment. Ionomycin saturates each sensor and provides a signal that is always higher than the histamine peak response, confirming that none of the sensors was saturated by histamine. See Supplementary Table ST1 for details on statistical tests and sample sizes.

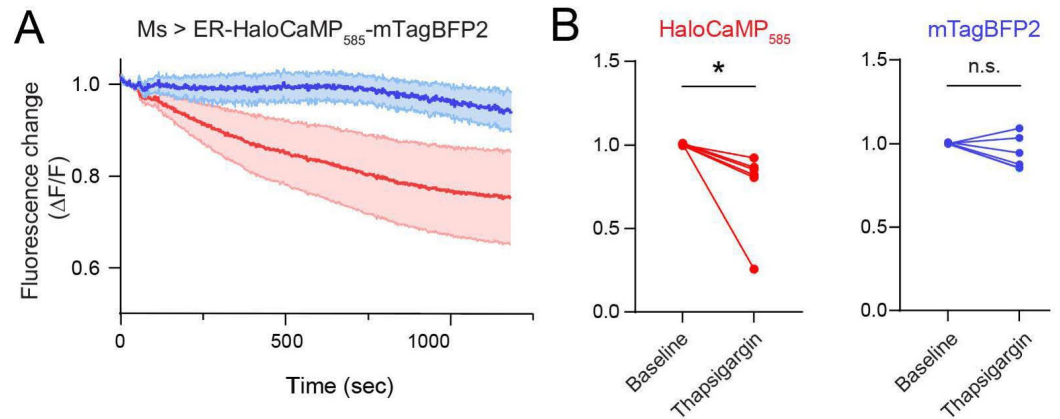


Fig. S4.

Simultaneous measurements of ER-HaloCaMP₅₈₅ and mTagBFP2 in Ms fly neurons expressing ER-HaloCaMP₅₈₅

(A) Mean relative fluorescence changes normalized to baseline over time in response to ER calcium depletion induced by thapsigargin in ER-HaloCaMP₅₈₅ and mTagBFP2. Shaded areas represent SEM. **(B)** Quantification of fluorescence changes for ER-HaloCaMP₅₈₅ (left, red) and mTagBFP2 (right, blue) at baseline and after thapsigargin application in paired samples. ER-HaloCaMP₅₈₅ fluorescence significantly decreased whereas mTagBFP2 fluorescence remained stable (n.s., not significant). See Supplementary Table ST1 for details on statistical tests and sample sizes

Acknowledgements

We would like to thank all members of the de Juan-Sanz laboratory for insightful discussions and comments. We thank Dr. Luke Lavis and the Janelia Open Science Initiative for providing the Janelia Fluor dyes used in this study. Part of this work was carried out in the ICM.Quant core facility (RRID:SCR_026393). We gratefully acknowledge Claire Lovo, Meryem Rezzik and Astou Tangara for setting up multiplexing experiments. Animals were hosted with the help of the PHENOPARC facility at ICM. This work was made possible by the Paris Brain Institute Diane Barriere Chair in Synaptic Bioenergetics awarded to J.d.J.-S., funding from the French national program “Investissements d’avenir” ANR-10-IAIHU-0006 awarded to the ICM and funding by the Richard Mille Fund (Project DBS, 2023-2028). Our additional funding sources are an ERC Starting Grant SynaptoEnergy (European Research Council, ERC-StG-852873), 2019 ATIP-Avenir Grant (CNRS, Inserm), and two grants by the French National Research Agency (ANR) under project numbers ANR-24-CE16-0221 and ANR-22-CE16-0020 awarded to J.d.J.-S. D.H. is funded by an ERC Starting Grant GutSense (European Research Council, ERC-StG-101117267). A.M. is the recipient of a predoctoral fellowship from the French Ministry of Science. J.d.J.-S. is a permanent CNRS researcher and a FENS-Kavli Scholar. V.R. is supported by the Max Planck Society.

Additional information

Author contributions

Conceptualization, J.d.J.-S.; methodology, A.M., H.F., R.F., K.Z., B.S., C.R., J.d.J.-S.; investigation, A.M., H.F., R.F., V. R., B.S., C.R., K.Z., J.d.J.-S.; project administration and supervision, J.d.J.-S.; writing - original draft, A.M. and J.d.J.-S.; writing - review & editing, A.M., H.F., R.F., K.Z., B.S., C.R., C.G., T.O., D.H., V.R., E. R .S. and J.d.J.S.

Additional files

Supp. Table S1 [↗](#)

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Reviewer #1 (Public review):

Summary:

The manuscript by Moret et al. details the development and characterisation of novel ER- and mitochondria-targeted genetically encoded chemogenic Ca²⁺ sensors.

Strengths:

Compared to existing probes, these sensors exhibited superior responsiveness, brightness, and photostability within the red and far-red emission spectrum, enabling triple compartment Ca²⁺ measurements (ER, mitochondria, cytosol) and the detection of Ca²⁺ dynamics in axons and dendrites.

Weaknesses:

The data are robust and convincing, although the manuscript text lacks precision.

<https://doi.org/10.7554/eLife.107845.1.sa1>

Reviewer #2 (Public review):

Summary:

Moret et al. present an engineered family of fluorescent calcium indicators based on HaloCamp, a HaloTag-based sensor system that utilizes Janelia Fluorophores (JF dyes) to report calcium dynamics. By introducing single or multiple amino acid substitutions, the authors reduce HaloCamp's calcium affinity, making these low-affinity variants well-suited for imaging calcium transients in high-calcium environments such as the endoplasmic reticulum (ER) and mitochondria. The study validates the sensors' dissociation constants (K_d), spectra, and multiplex capabilities. It demonstrates improved performance compared to existing tools when targeted to subcellular compartments in mammalian cells and cultured neurons. The sensors can be tuned across the red-to-far-red spectrum via JF585 and JF635 labeling, enabling flexible multiplexed imaging. For example, the authors show that HaloCamp can be targeted to mitochondria and used alongside other green and red sensors, allowing simultaneous imaging of calcium dynamics in the cytosol, ER, and mitochondria. Overall, they achieve their goals, and the data demonstrate that HaloCamp variants are effective for detecting ER and mitochondrial calcium changes under physiological conditions. The presented experiments support the conclusions. However, some key aspects, such as sensor kinetics and axonal validation, would benefit from further analysis.

This work is likely to have an important impact on the fields of calcium imaging and organelle physiology. The modular design of HaloCamp and its compatibility with a wide range of fluorophores offer a broad application range for cell biologists and neuroscientists.

Strengths:

- (1) The authors introduce the first tunable, dye-based, low-affinity HaloTag calcium sensors for subcellular imaging, addressing a significant unmet need for ER and mitochondrial calcium detection.
- (2) The ability to pair HaloCamp with JF585 and JF635 extends the spectral range, facilitating multiplexed imaging with existing calcium indicators.
- (3) The sensors are validated in a range of subcellular compartments (ER, mitochondria, cytosol) in both mammalian cells and neurons.
- (4) The authors successfully demonstrate simultaneous imaging of three compartments using orthogonal sensors, a technically impressive feat.
- (5) K_d values are measured, and fluorescent responses are tested under physiologically relevant stimulation.

Weaknesses:

- (1) The authors do not quantify the kinetics (e.g., decay tau or off-rate) of the fluorescent signals, particularly after stimulation. For example, in the ER imaging experiments in neurons, the decay of the HaloCamp fluorescence after field stimulation (20 APs @ 20 Hz) is not analyzed or compared to ER-GCaMP6-210 or R-CEPIer.
- (2) It remains unclear whether the observed decay represents the sensor's off-kinetics or actual physiological calcium clearance from the ER. A comparison between sensors or an independent measurement of ER clearance rates in vitro would clarify this.
- (3) The choice of 20 APs at 20 Hz is not justified. Specifically, single APs or low-frequency stimulations are not tested, leaving unclear what the detection threshold of the new sensors is.
- (4) In neuron experiments, the authors report measuring ER calcium in axons based presumably on morphology, but no specific justification for selection, markers, or post hoc labeling is described.
- (5) Figure 5 assumes that all three indicators (cytosolic, ER, and mitochondrial) are fast enough to report calcium dynamics in response to histamine. This assumption is not fully validated. Cross-controls (e.g., expressing GCaMP6-210 in mitochondria and HaloCamp in the ER) would strengthen confidence that the sensors are correctly reporting dynamic changes.
- (6) It is not clear why Thapsigargin leads to depletion in HeLa cells and neurons in experiments shown in Figure 1E, but not in 2B upon field stimulation.

<https://doi.org/10.7554/eLife.107845.1.sa0>